

\$PEACE

Peace Soldiers White Paper

Engineers and Inverstors version

Final draft

November 2025

Table of Contents

\$PEACE	1
Table of Contents	2
Chapter 1 – Executive Summary	8
1.1 Project Overview	8
1.2 Executive Summary	8
1.3 Key Findings	9
1.4 Methodology	9
1.5 Deep-Dive Introduction	9
1.6 Risk Matrix	10
1.7 Creative Scenarios	10
1.8 Founders’ Statement — Principles & Policies	11
1.9 Message to Investors and DAO Members	11
1.10 How to Navigate This Whitepaper	11
Chapter 2 — Vision, Mission, and Strategic North Star	11
2.1 Executive Summary	11
2.2 Key Findings	12
2.3 Methodology	12
2.4 Deep-Dive Analysis	12
2.4.1 The Vision	12
2.4.2 Strategic Pillars and Objectives	13
2.4.3 Why Solana + Telegram now	14
2.4.4 External Signals Shaping the Vision	14
2.5 Risk Matrix	15
2.6 Creative Scenarios	15
2.7 Vision-Aligned Success Metrics (first 12 months)	16
Chapter 3 — Problem & Market Need	17
3.1 Executive Summary	17
3.2 Key Findings	17
3.3 Methodology	17
3.4 Deep-Dive Analysis	18
3.4.1 The Structural Problem: “Broken Trust Supply Chain”	18
3.4.2 Demand Signals & Macro Context	18
3.4.3 Trust & Fraud Reality Check	18
3.4.4 Why the Timing Is Right (Rails & UX)	18
3.4.5 Market Map (Who Feels the Pain)	19
3.4.6 From Problem to Requirements (Implications)	19
3.5 Risk Matrix	20
3.6 Creative Scenarios	20
3.7 Market Size & Addressability (Top-Down context)	21

3.8 Original Framework — “Trust Supply Chain Index (TSCI)”	21
Chapter 4 — Solution & Architecture	22
4.1 Executive Summary	22
4.2 Key Findings	22
4.3 Methodology	22
4.4 Architecture Overview (Systems Map)	23
4.5 Component Deep-Dive	25
4.5.1 Identity: Peace Passport cNFT	25
4.5.2 Missions & Verification	25
4.5.3 Rewards, Burn, and Budget Discipline	26
4.5.4 Peace Store & Anti-Wash Commerce	26
4.5.5 Governance & Config Registry	26
4.5.6 Data Plane & Public KPIs	27
4.6 On-Chain & Off-Chain Research Plan	27
4.7 Security Model & Threat Mitigations	28
4.8 Creative Framework — “The Parametric City”	29
4.9 Rollout Plan	29
4.10 Quality Gate	30
Chapter 5 — Technical Specifications	31
5.1 Executive Summary	31
5.2 Key Findings	31
5.3 Methodology	31
5.4 Program Suite & Responsibilities	32
5.4.1 Identity Program (passport)	32
5.4.2 Missions Program (missions)	32
5.4.3 Rewards Program (rewards)	33
5.4.4 Treasury & Buyback Program (treasury)	34
5.4.5 Store Program (store)	35
5.4.6 Governance Program (dao)	35
5.4.7 Config Registry Program (config)	36
5.5 Off-Chain Perimeter & Data Plane	36
5.5.1 API Gateway	36
5.5.2 Verification Pipeline	37
5.5.3 Indexer & Analytics	37
5.6 Data Schemas (selected)	37
5.7 Security, Gas/Compute, and Rate Limits	38
5.8 Testing & Audit Plan	39
5.9 Operational SLAs & Telemetry	39
5.10 Creative Analogy — “Air Traffic Control for Impact”	40
5.11 Rollout Checklist (Engineer’s Cut)	40
Chapter 6 — Tokenomics	42

6.1 Executive Summary	42
6.2 Key Findings	42
6.3 Methodology	42
6.4 Deep-Dive Analysis	43
6.4.1 Supply, Issuance, and Burn	43
6.4.2 Rewards Budget & Coverage	43
6.4.3 Buyback & Burn — Profit-Conditioned	43
6.4.4 Token Utility (Demand Surfaces)	44
6.4.5 Distribution & Vesting (Principles)	44
6.4.6 Emission & Velocity Dynamics	44
6.4.7 Economic Stress Tests	44
6.5 Risk Matrix	45
6.6 Creative Scenarios	46
6.7 Metrics & Telemetry	46
6.8 Quality Gate	47
Chapter 7 — Governance	48
7.1 Executive Summary	48
7.2 Key Findings	48
7.3 Methodology	48
7.4 Governance Topology	49
7.4.1 Participants & Roles	49
7.4.2 Proposal Classes	49
7.4.3 Voting Mechanics	49
7.4.4 Timelocks & Execution	50
7.5 What Governance Controls (Scope)	50
7.6 Integrity & Safety Rails	51
7.6.1 Capture Resistance	51
7.6.2 Proposal Hygiene	51
7.6.3 Treasury Guardrails	51
7.6.4 Emergency Response (SOP)	52
7.7 Governance Data Plane	52
7.8 Governance Processes (Playbooks)	52
7.8.1 Parameter Change (Economic)	52
7.8.2 Treasury Buyback Adjustment (Critical)	52
7.8.3 EmergencyStop Engage/Release	53
7.8.4 Grants / NGO Budget Allocation	53
7.9 Risk Matrix	53
7.10 Creative Framework — “The Three-Chamber Heart”	54
7.11 Progressive Decentralization (Roadmap)	54
7.12 Quality Gate	54
Chapter 8 — Human Impact Index (HII) & Missions	56

8.1 Executive Summary	56
8.2 Key Findings	56
8.3 Methodology	56
8.4 HII Design — The Function	57
8.5 Missions — From Template to Verdict	58
8.5.1 Mission Templates (E / V / A)	58
8.5.2 Verification Pipeline	58
8.5.3 Disputes & Penalties	58
8.6 Anti-Fraud: Layers & Costs	59
8.7 HII → Rights, Rewards, and Utility	59
8.8 On-Chain & Off-Chain Research Plan	60
8.9 Fairness & Accessibility	60
8.10 Risk Matrix	61
8.11 Creative Framework — “The Impact Altimeter”	61
8.12 Quality Gate	62
Chapter 9 — Peace Store & Commerce	63
9.1 Executive Summary	63
9.2 Key Findings	63
9.3 Methodology	63
9.4 Commercial Model	64
9.4.1 Roles and Incentives	64
9.4.2 Fee & Discount Surfaces	64
9.4.3 Clean Revenue Logic	64
9.5 Merchant Lifecycle	65
9.6 Anti-Wash & Risk Controls	66
9.6.1 Detection Primitives	66
9.6.2 Policy Levers	66
9.6.3 Dispute & Appeals	66
9.7 KPIs & Definitions (Publish by Default)	66
9.8 Economic Scenarios	67
9.9 User Experience (Funnel, Not Friction)	68
9.10 Compliance & Category Governance	68
9.11 Risk Matrix	69
9.12 Creative Framework — “The Harbor Model”	69
9.13 Quality Gate	70
Chapter 10 — Risk, Legal, and Privacy	71
10.1 Executive Summary	71
10.2 Key Findings	71
10.3 Methodology	72
10.4 Deep-Dive: Risk Domains & Controls	72
10.4.1 Market & Treasury Risk	72

10.4.2 Governance & Conduct Risk	72
10.4.3 Operational & Fraud Risk	73
10.4.4 Counterparty & Merchant Risk	73
10.4.5 Privacy & Data Protection	73
10.4.6 Compliance & Financial Crime	74
10.4.7 Consumer & Advertising Risk	74
10.4.8 IP & Content Policy	74
10.5 Compliance Matrix (What, How, Evidence)	75
10.6 Privacy Architecture (PII-Sparse by Design)	76
10.7 Incident Response & Business Continuity	76
10.8 Disclosures & Document Set (Versioned)	76
10.9 Risk Matrix (Consolidated)	77
10.10 Creative Framework — “Compliance as a Living Ledger”	78
10.11 Quality Gate	78
Chapter 11 — Roadmap & Delivery	79
11.1 Executive Summary	79
11.2 Key Findings	79
11.3 Methodology	79
11.4 Phased Roadmap (Milestones, Owners, Evidence)	80
Phase 0 — Bootstrap (Weeks 0–4)	80
Phase 1 — Closed Beta (Weeks 5–10)	80
Phase 2 — Open Beta (Weeks 11–20)	81
Phase 3 — Global Launch (Weeks 21–32)	81
Phase 4 — Progressive Decentralization (Weeks 33+)	81
11.5 Workstreams, Deliverables, and Acceptance	82
11.6 Dependencies & Critical Paths	83
11.7 Team Structure, RACI, and Budget Envelopes	83
11.8 OKRs (Quarterly)	84
11.9 QA & Launch Readiness (LTR)	84
11.10 Communications Plan	85
11.11 Risks to the Plan & Mitigations	85
11.12 Creative Framework — “Dual-Lane Highway”	86
11.13 Quality Gate	86
Chapter 12 — Team & Organization	87
12.1 Executive Summary	87
12.2 Key Findings	87
12.3 Methodology	87
12.4 Org Topology (Pods, Councils, and Lines)	88
12.4.1 Product Pods (P&L-adjacent)	88
12.4.2 Risk & Policy Councils (Guardrail owners)	88
12.5 Decision Rights & Escalation	89

12.6 Role Charters (Critical Roles)	90
12.7 Hiring Plan (First 9–12 Months)	90
12.8 Operating Cadence	91
12.9 Performance System (OKRs & Promotions)	91
12.10 Security & Privacy Operations	92
12.11 Vendor & Partner Governance	92
12.12 Community & Support	93
12.14 Risk Matrix (Org-centric)	93
12.15 Creative Framework — “The Orchestra Model”	94
12.16 Quality Gate	95
Chapter 13 — Annexes (Parameter Registry, SOPs, KPI Specifications)	96
Annex A — On-Chain Config Registry (Canonical Keys)	96
Annex B — Parameter Defaults (v1 Baseline)	100
Annex C — Governance Quorums, Timelocks, Classes	100
Annex D — SOPs (Runbooks)	101
Annex E — Data Schemas & KPI Specifications	102
Annex F — Threat Model (Summary)	104
Annex G — Document Versions (External-Facing)	105
Annex H — Glossary (Selected)	105
End Matter — Consolidated Actionable Recommendations & Appendices	106
EM-1 — Consolidated Actionable Recommendations	106
EM-2 — Appendices (Chapter-Scoped Reference Packs)	108
Peace Constellation (Positioning)	110
Impact Gravity Index (IGI)	110
Governance Health Score (GHS)	110
Risk Absorption Ratio (RAR)	111
Verification Pipeline — Signals & Weights	111
Figures (All figures will be injected into the final version)	111
References	115

Chapter 1 – Executive Summary

1.1 Project Overview

Peace Soldiers is a human-centric Web3 ecosystem that turns real-world pro-social actions into verifiable digital value and long-term community wealth. The stack includes: a Telegram MiniApp for Web2-grade access, a soulbound Peace Passport cNFT for identity, a 10,000-piece Peace Soldiers NFT collection for loyalty and access, and a fixed-supply \$PEACE token that powers incentives, governance, and the Peace Store economy. Together, these modules create a closed-loop flywheel—impact → verification → rewards → governance → treasury discipline—built on Solana [R1][R1a] for performance and low costs.

1.2 Executive Summary

The problem: Traditional impact channels face weak transparency, short-lived incentives, and unreliable measurement. Many Web3 attempts stalled at pilot scale due to UX friction and token misalignment.

The solution:

- **Identity & Reputation:** Peace Passport (soulbound cNFT) provides unique, non-transferable identity as a base layer for Sybil resistance and reputation.
- **Action → Value:** The Human Impact Index (HII) scores learning, volunteering, and activity with multi-signal verification; HII directly influences rewards, governance weight, and store benefits.
- **Economy:** Fixed-supply \$PEACE with two discipline levers—claim-time issuance burn and buyback-and-burn contingent on net treasury profits.
- **Governance:** Formal proposal → vote → timelock → execute flow, anti-capture rules, and progressive decentralization tied to real readiness metrics.
- **Commerce:** Peace Store marketplace with anti-wash controls and on-chain parameterization.

Why now: Verified outcomes, usable UX, and sound token engineering make impact operable, measurable, and capital-efficient—so contributions compound rather than fade.

1.3 Key Findings

1. **Audit-ready by design.** Each module maps to a discrete control surface—identity, mission proofs, rewards budget, store policy, buyback SOPs, and DAO thresholds—with verifiable change logs.
2. **Aligned incentives with credible deflation.** Dual discipline (claim-time burn + net-profit buybacks) reduces extractive behavior and targets durability.
3. **Resilient governance.** Vote caps per identity, HII-weighted participation, quorum classes, and timelocks add friction against manipulation.
4. **Impact as utility.** HII affects staking, governance weight, and store benefits—turning “doing good” into tangible economic surface area.

1.4 Methodology

- **Source synthesis:** Consolidated architecture, tokenomics, governance, and HII/mission specifications into a single executive narrative.
- **Design-for-audit:** Emphasized on-chain configuration, SOP discipline for buybacks/treasury/emergencies, and transparent reporting.
- **Risk triangulation:** Highlighted anti-fraud verification and governance security drawn from the system’s control surfaces.

1.5 Deep-Dive Introduction

The Flywheel (Impact → Value → Voice → Stability):

- **Impact:** Users complete verified missions; signals include GPS/device data, time-stamped media, host signatures, and optional peer review.
- **Value:** Rewards accrue in \$PEACE with a DAO-tunable claim-time burn to temper issuance and align with sustainable supply.
- **Voice:** HII boosts voting weight and unlocks higher-tier missions; caps and quorums reduce plutocratic risk.
- **Stability:** Treasury buybacks (TWAP-guarded, oracle freshness checks, circuit breakers) recycle net profits into token support with public reporting. (See Annex E: Oracle Providers & Max-Age table.)

Original Analogy — “Impact Nervous System”:

Picture the ecosystem as a **neural network**. Missions are sensory inputs; HII updates synaptic weights; \$PEACE rewards are neurotransmitters; governance is the prefrontal cortex applying policy; buyback & burn maintain homeostasis. Over time, the system “learns”—reinforcing verified impact and pruning noise.

1.6 Risk Matrix

Risk	Vector	Mitigation	Control Surface
Governance capture	Vote concentration, flash mobilization	Vote caps per ID, HII weight, timelocks, progressive decentralization	DAO thresholds & SOPs
Token pressure	Excess issuance, macro shocks	Claim-time burn; TWAP-guarded buybacks; transparency dashboards	Tokenomics & Buyback SOP
Mission fraud	GPS spoofing, fabricated media	Multi-signal verification, host signatures, random audits, penalties/appeals	Anti-Fraud & Verification
Store abuse	Wash-trading, merchant risk	Anti-wash policy, KYC tiers, risk scoring, fee splits, dispute resolution	Store module & Config Registry

1.7 Creative Scenarios

1. **Regulatory “Tectonic Shift”:** If policy changes abruptly (e.g., sanctions, KYC thresholds), parameters move like plates along a fault line—timelocks and registry updates absorb shocks without breaking continuity.
2. **Liquidity Drought:** In risk-off regimes, buybacks never trigger without net profit—protecting treasury while claim-burn sustains supply discipline.
3. **HII Surge:** Rapid growth in verified actions raises HII-weighted voting and store discounts—managed via budget ceilings and tunable caps to keep incentives solvent.

1.8 Founders' Statement — Principles & Policies

- **Utility-first token:** no profit-sharing promises; value accrues through **use**, **access**, and **governance** under applicable laws.
- **Transparency and restraint:** clear vesting, hard locks for team/partners, and visible circulating-supply dashboards.
- **Human dignity as a design constraint:** data minimization; on-chain hashing with compliant off-chain erasure upon request.

1.9 Message to Investors and DAO Members

This is a **measured system** where verified human actions, governed parameters, and treasury discipline converge. You'll see the levers, the guardrails, and the reports—by design. Join to build a **scaled peace economy** where doing good translates into durable digital capital.

1.10 How to Navigate This Whitepaper

The canonical ticker across the stack is \$PEACE. Any legacy variants are aliases only and will be sunset in the next doc refresh.

Brand/Ticker Freeze Note

Problem & Market Need → Solution & Architecture → Tokenomics → Governance → HII & Missions → Store & ReFi → Risk/Legal → Roadmap → Team → Annexes → **End Matter.**

Chapter 2 — Vision, Mission, and Strategic North Star

2.1 Executive Summary

Peace Soldiers exists to turn verified human contribution into durable digital capital. The vision is a global peace economy where identity, action, and value compound: people complete real missions, proofs are verified, reputation grows, and rewards unlock access, governance, and commerce. The mission: ship a frictionless, audit-ready impact stack—Telegram [R2][R3][R4] MiniApp onboarding, Peace Passport (soulbound cNFT) identity, Human Impact Index (HII), a fixed-supply \$PEACE token with discipline levers, and an anti-wash Peace Store—so that doing good is as easy as sending a message and as measurable as a ledger entry. This chapter defines the destination, the strategic lanes, and the tests of success.

2.2 Key Findings

1. **North-Star clarity:** The ecosystem's purpose is not charity; it's **unit-economics-positive impact**—where verified actions feed revenue flywheels (Store, partnerships) and disciplined token mechanics (claim-burn + buyback on net profit) stabilize value over time.
2. **Friction kills impact; MiniApps solve it:** Telegram MiniApps offer native-like flows, payments, and notifications in the chat context users already inhabit—reducing the adoption tax that stalls most impact apps.
3. **Identity must be non-transferable:** A soulbound Peace Passport makes reputation sticky and Sybil-resistant, enabling fair rewards and sane governance.
4. **Design for reputational and economic resilience:** Governance guardrails (vote caps per ID, HII weight, timelocks) and token discipline protect against capture and inflationary drift while maintaining usability.
5. **Narrative ≠ hype:** The vision wins if—and only if—metrics show **compounding verified actions, repeat merchants, and stable reward coverage**.

2.3 Methodology

- **Vision extraction:** Distilled the project’s stated aims into measurable outcomes and operating principles; mapped each principle to a concrete control surface (identity, verification, incentives, governance, treasury, commerce).
- **Comparative scan:** Benchmarked UX, identity, and adoption hypotheses against public docs for **Telegram MiniApps**, **soulbound tokens literature**, **Solana costs/fees**, and **global adoption trends**.
- **Risk lens:** Integrated 2024–2025 fraud and adoption signals to align the vision with realistic threat models.

2.4 Deep-Dive Analysis

2.4.1 The Vision

A **scaled peace economy** where a billion micro-actions are **verifiably recorded**, **equitably rewarded**, and **productively spent**—creating a positive-sum loop among citizens, NGOs, merchants, and the treasury. The end state is not a one-time campaign; it’s a **self-healing network** that learns which actions produce enduring social good and allocates capital accordingly.

Original framework — “Constellation Strategy”

Think of the ecosystem as a **constellation**:

- **Star A: Access (MiniApp).** Low-friction, device-native flows (authorization, payments, notifications) so users can act within seconds.
- **Star B: Identity (Peace Passport cNFT).** Non-transferable identity anchors provenance, reputation, and Sybil resistance.
- **Star C: Proof (HII Engine).** Multi-signal verification (GPS/media/host/peer) translates actions into a robust score.
- **Star D: Value (\$PEACE).** Fixed supply with discipline levers—claim-time issuance burn and **TWAP-guarded** buybacks funded only by net profits.
- **Star E: Commerce (Peace Store).** Anti-wash marketplace, merchant onboarding, and reward-linked discounts.

- **Star F: Voice (DAO).** Vote caps per ID, HII weighting, quorums, and timelocks to keep power broad and decisions paced.

As with real constellations, **navigation** happens by connecting multiple points at once: progress requires *all* stars to be visible and properly aligned.

2.4.2 Strategic Pillars and Objectives

- **Pillar 1 — Frictionless participation.**
Objective: 1-click identity, <30s mission start time, and integrated fiat/crypto payments via MiniApps. **KPI:** Mission start-to-submit $\leq$ 3 taps median; Day-30 retention $\geq$ 30%.
- **Pillar 2 — Verifiable action at scale.**
Objective: Anti-fraud signals and randomized audits; false-positive rate < 2%. **KPI:** Verified-mission share $\geq$ 90%; appeal resolution $\leq$ 72h.
- **Pillar 3 — Incentive solvency.**
Objective: Reward budget coverage $\geq$ 12 months at current run-rate; claim-burn calibrated to issuance drift; buybacks limited to realized net profit with transparent logs. **KPI:** Coverage ratio, Buyback SOP compliance rate.
- **Pillar 4 — Merchant gravity.**
Objective: Repeat-merchant rate $\geq$ 60%; GMV growth with anti-wash filters. **KPI:** Store GMV (de-washed), discount elasticity vs. HII tier.
- **Pillar 5 — Resilient governance.**
Objective: No single entity > X% effective voting power post-cap/HII; proposal timelock $\geq$ Y hours; emergency SOP tested quarterly. **KPI:** Gini/Herfindahl of voting power; audit log completeness.

2.4.3 Why Solana + Telegram now

- **Throughput & fees:** Solana's base and prioritization fee model allows low, predictable costs at consumer scale (sub-cent execution typical), essential for mission submissions and micro-rewards.
- **Distribution:** Telegram MiniApps are available across Android, iOS, macOS, desktop, and web variants; full-screen UX and native payments simplify mainstream onboarding.

2.4.4 External Signals Shaping the Vision

- **Adoption shifts:** Chainalysis reports continued grassroots adoption across emerging markets, implying real demand for low-cost rails and practical UX.

- **Fraud environment:** Scam sophistication remains elevated (GenAI-assisted social engineering), reinforcing the need for strong verification and governance throttles.
- **ReFi tailwinds:** Regenerative-finance reports show growing appetite for market-linked impact products—supporting the Store and partnerships thesis.

2.5 Risk Matrix

Risk	Description	Likelihood	Impact	Primary Mitigations
Adoption stall	Onboarding friction, unclear value	Med	High	MiniApp native flows; clear HII → benefit mapping; progressive disclosure UX. Telegram
Sybil & farming	Fake identities, scripted missions	Med	High	Soulbound Passport; multi-signal proofs; random audits; penalties & appeals. SSRN
Token pressure	Over-issuance or macro drawdowns	Med	High	Claim-time burn, buybacks only on net profit, TWAP guards, transparency.
Merchant wash	Fake GMV & coupon abuse	Med	Med	Anti-wash filters, risk scoring, tiered KYC, dispute process.
Regulatory shift	KYC/AML or sanctions changes	Low-Med	High	Parameterized compliance toggles, timelock updates, jurisdictional ring-fencing.
Narrative risk	“Impact” perceived as vague	Med	Med	Public KPIs (verified missions, repeat merchants, coverage ratio), monthly reports.

2.6 Creative Scenarios

1. **“Constellation Drift” (Adoption shock):** A viral campaign drives 10× mission volume in a week. **Response:** Raise verification sampling, temporarily lower per-user caps, and publish a mid-cycle KPI note. The “stars” stay aligned by increasing proof gravity before tuning rewards.
2. **“Gravity Well” (Merchant-led pull):** A marquee partner joins the Peace Store, spiking GMV. **Response:** Activate anti-wash thresholds, tighten discount elasticity for high-HII tiers, and use net profits to schedule TWAP buybacks within SOP windows—only if

coverage remains >12 months.

3. **“Solar Flare” (Regulatory event):** A new KYC rule lands. **Response:** Activate jurisdictional toggle in Config; freeze affected tiers; queue DAO update with timelock; maintain continuity for other regions.

2.7 Vision-Aligned Success Metrics (first 12 months)

- **Participation:** 500k MiniApp MAU; $\geq 35\%$ 30-day retention; $\geq 65\%$ mission completion rate.
- **Proof Quality:** $\geq 90\%$ verified-mission share; $< 2\%$ false positives; appeal turnaround $\leq 72\text{h}$.
- **Economy:** Reward budget coverage ≥ 12 months; TVDR (Token Velocity Decay Rate) trending downward QoQ; Buyback SOP adherence 100%.
- **Commerce:** 1,000 active merchants; repeat-merchant rate $\geq 60\%$; de-washed GMV growth $\geq 20\%$ MoM.
- **Governance:** Effective voting power of largest entity $\leq X\%$; proposal timelock $\geq Y$ hours; quarterly emergency-drill pass rate 100%.
- **Trust:** Monthly KPI + treasury reports on-chain (hash) and web; uptime SLA $\geq 99.5\%$.

Chapter 3 — Problem & Market Need

3.1 Executive Summary

Billions of dollars in pro-social effort fail to convert into sustained outcomes because the **trust supply chain** is broken: identities are duplicable, proofs are weak, incentives decay, and reporting is opaque. At the same time, consumer adoption of crypto rails is **rising in the very regions** where low-cost, high-throughput, mobile-first tools matter most—creating a window to rebuild the stack with verifiable identity, credible proofs, disciplined token mechanics, and mainstream UX (Telegram MiniApps + Solana).

3.2 Key Findings

1. **Demand is real; rails are ready.** Global grassroots crypto adoption remains strongest in emerging markets—where mobile-native, low-fee rails can unlock real use, not speculation.
2. **Impact capital is supply-constrained and misallocated.** The SDG financing gap runs into **trillions per year**, and public debt burdens squeeze social budgets—raising the bar for transparent, capital-efficient impact flows.
3. **Trust failures are costly.** Fraud and internal control weaknesses are persistent in the nonprofit/charity sector, undermining donor confidence and wasting scarce resources.
4. **UX is the adoption lever.** MiniApps now run full-screen with native-like flows, authorization, and payments—reducing the “download a new app” tax that kills funnels.
5. **Unit economics matter.** Solana’s sub-cent base fees and optional priority fees keep mission submissions and micro-rewards economically viable at scale.

3.3 Methodology

- **Market scan:** Synthesized reputable external sources on: global crypto adoption (Chainalysis), SDG financing gaps (UN/FSDR), debt overhang (AP), charity fraud studies (BDO/MHA), Solana fee model, and Telegram MiniApp capability updates.
- **Problem interviews (proxy):** Derived pain points from public sector, NGO, and ReFi literature; mapped them to concrete control surfaces (identity, verification, incentives, governance, commerce).

- **On-chain lens:** Defined baseline metrics to monitor adoption and integrity once live (e.g., **Mission Verification Rate**, **De-Washed GMV**, **Token Velocity Decay Rate**), to be tracked on a public dashboard.

3.4 Deep-Dive Analysis

3.4.1 The Structural Problem: “Broken Trust Supply Chain”

Most impact systems leak value at four links:

1. **Identity:** Reusable or fake personas enable Sybil/farming;
 2. **Proof:** Low-entropy evidence (photos, check-ins) is easy to forge;
 3. **Incentive:** Rewards are front-loaded, poorly budgeted, and farmable;
 4. **Accounting:** Reporting lags and lacks independent, real-time verification.
- These leaks depress donor/merchant confidence and push the ecosystem into **short-termism**.

Why this persists:

- **Measurement costs** outstrip the size of many micro-actions, so verification is skipped or under-resourced.
- **Legacy rails** impose high transaction and coordination costs, blocking micro-rewards and small grants.
- **Opaque reporting** makes it hard to separate genuine impact from marketing.

3.4.2 Demand Signals & Macro Context

- **Grassroots crypto adoption:** Chainalysis’ 2024 index shows sustained strength across Central & Southern Asia and other emerging regions, indicating real user maturity on cheap, mobile rails.
- **SDG financing gap:** UN’s 2024 FSDR and related analyses cite a **multi-trillion-dollar** annual shortfall—private-sector participation and efficient channels are essential.
- **Debt overhang:** Record public debt pressures social spending in developing countries, increasing reliance on efficient, transparent private financing and community-driven models.

3.4.3 Trust & Fraud Reality Check

- **Charity fraud prevalence:** Sector surveys report material fraud exposure, including **insider fraud**—a key reason donors demand verifiable, tamper-evident reporting.

3.4.4 Why the Timing Is Right (Rails & UX)

- **Rails:** Solana's fee model (base ~5,000 lamports; optional priority fees) makes high-frequency, micro-value interactions feasible.
- **UX:** Telegram MiniApps deliver full-screen, app-like experiences with built-in auth, notifications, and payment support—compressing onboarding friction.

3.4.5 Market Map (Who Feels the Pain)

- **Citizens/Volunteers:** Want reputation and meaningful rewards without complex wallets or app installs.
- **NGOs/Hosts:** Need low-cost verification and predictable, auditable reward programs.
- **Merchants:** Want incremental, measurable demand (discounts linked to verified reputation), not coupon arbitrage.
- **Donors/Partners:** Need transparent, tamper-evident flows and KPI dashboards.
- **Regulators/Exchanges:** Want evidence of compliance-ready identity and controls.

3.4.6 From Problem to Requirements (Implications)

- **Non-transferable identity** to anchor reputation and cap voting power per human.
- **Multi-signal verification** (GPS, time-stamped media, host attestations, randomized audits) to raise the cost of fraud.
- **Budget-disciplined incentives** (claim-time burn, buybacks only on realized net profit) to prevent runaway issuance.
- **Anti-wash commerce** and **public KPIs** to protect merchants and donors and surface real outcomes.

3.5 Risk Matrix

Risk	Description	Likelihood	Impact	Why it matters now
Adoption friction	Extra apps/wallets reduce conversion	Medium	High	MiniApps compress steps; mobile-native rails align with emerging-market habits.
Sybil/farming	Fake or duplicate identities	Medium	High	Soulbound identity + multi-signal proofs raise attack cost; random audits deter.
Fraud & misreporting	Insider/external fraud in hosts/NGOs	Medium	High	Sector studies show persistent fraud; real-time, on-chain evidence is required.
Token inflation pressure	Over-issuance vs. sustainable demand	Medium	High	Claim-burn + buybacks (net-profit only) and budgets keep incentives solvent.
Regulatory shifts	New KYC/AML or sanctions rules	Low–Med	High	Parameterized toggles and timelocks reduce disruption; jurisdictional scoping.
Narrative fatigue	“Impact” dismissed as marketing	Medium	Medium	Public KPIs (de-washed GMV, verification rate, TVDR) build evidence over time.

3.6 Creative Scenarios

1. **“Leaky Pipe” Stress Test:** Rapid user growth with weak verification tooling causes leakages (fraud attempts, coupon abuse). **Response:** ratchet verification sampling, cap per-user rewards temporarily, and publish a live integrity score until tooling catches up.
2. **“Remittance Crossover”:** Merchants in a corridor with high crypto usage start accepting discounts in-app; GMV spikes as users spend rewards locally. **Response:** enable corridor-specific fee splits and highlight de-washed GMV in public KPIs to prove real commerce.

3. **“Regulatory Plate Shift”**: A jurisdiction tightens KYC for reward payouts. **Response**: activate compliance tier for that region via config, queue DAO update with timelock, and maintain operations elsewhere with unchanged UX.
4. **“Liquidity Winter”**: Macro drawdown pressures token price. **Response**: buyback remains off without realized net profit; instead, widen claim-burn band modestly and prioritize Store partnerships to add non-speculative demand.

3.7 Market Size & Addressability (Top-Down context)

- **SDG finance gap: \$2.5–\$4.0 trillion/year** gap suggests ample room for capital-efficient, transparent impact channels to complement public finance.
- **User base readiness**: Emerging-market crypto adoption and Telegram’s massive distribution (with MiniApp upgrades) indicate large reachable populations for mission-driven micro-transactions.
- **Unit economics feasibility**: Sub-cent base fees on Solana make micro-missions and micro-rewards economically sustainable versus legacy payment rails.

3.8 Original Framework — “Trust Supply Chain Index (TSCI)”

To quantify “market need” as a single number per region or partner, we propose **TSCI (0–100)**, combining:

- **Identity Integrity (30%)** — proxy from duplicate-ID attempts blocked / total signups.
- **Proof Strength (30%)** — verified-mission rate adjusted for audit sampling and false-positive rate.
- **Economic Soundness (25%)** — reward coverage months + TVDR trend.
- **Commerce Health (15%)** — de-washed GMV growth + repeat-merchant rate.
Regions with low TSCI are **high-need**: greater leakage today → greater ROI from deploying Peace Soldiers controls.

Chapter 4 — Solution & Architecture

4.1 Executive Summary

Peace Soldiers is engineered as a **modular, parameterized, and audit-ready** system that turns verified human actions into durable digital capital. The architecture aligns six domains—**Access, Identity, Proof, Incentives, Commerce, and Governance**—and stitches them with a **Config Registry, SOP playbooks**, and a **telemetry layer** so every change is observable, revertible, and attributable. The result is a flywheel that is fast enough for consumers, strict enough for auditors, and flexible enough for evolving regulation.

4.2 Key Findings

1. **End-to-end verifiability:** From cNFT identity issuance to mission proofs, from reward allocation to buyback execution, each step is **provable** and **parameterized** on-chain.
2. **UX without compromise:** Telegram MiniApp flows compress onboarding and mission submission into seconds while preserving security boundaries via signed payloads and guarded endpoints.
3. **Economic discipline by design:** Claim-time burn, buybacks that trigger only on **realized net profit**, and budget ceilings are embedded as contract-level constraints, not slideware.
4. **Governance safety rails:** Vote caps per identity, HII-weighting, quorums, and timelocks reduce capture risk; emergency SOPs and circuit breakers contain blast radius.
5. **Observability first:** Public KPIs, hashed snapshots (e.g., to IPFS/Arweave), and an indexer-based data plane enable third-party verification and replication.

4.3 Methodology

- **Requirements derivation:** Mapped Chapter 3 problem statements (identity leakage, weak proofs, incentive insolvency, opaque reporting) into concrete contract interfaces, registries, and off-chain services.

- **Threat modeling:** Identified primary attack classes (Sybil, proof forgery, wash trading, governance capture, oracle manipulation) and aligned them to layered mitigations.
- **Scalability targets:** Baked in throughput assumptions for high-frequency micro-interactions (mission posts, reward accruals, store purchases) on a low-fee L1.
- **Auditability pass:** Ensured every critical surface has (a) an immutable log, (b) configurable thresholds, (c) a deterministic SOP response.

4.4 Architecture Overview (Systems Map)

A. Access Layer (Client & Gateway)

- **Telegram MiniApp:** login/consent, mission wizard, wallet connect (custodial or self-custody), notifications.
- **API Gateway (stateless):** verifies signed requests, rate limits, and forwards to mission/identity services.
- **Content Service:** stores encrypted mission media off-chain with **on-chain hashes** to anchor provenance.

B. Identity Layer

- **Peace Passport (soulbound cNFT):** non-transferable identity with metadata pointers (e.g., commitment to minimal PII, jurisdiction flags).
- **ID Attestation Service:** optional KYC tiering; produces attestations that can gate reward tiers or merchant access.
- **Sybil Controls:** one-passport-per-human policy, device fingerprints (privacy-preserving), duplicate detection signals.

C. Proof & Scoring Layer

- **Mission Engine:** templates for E/V/A mission types (Education/Volunteering/Activity) with inputs, required signals (GPS, media watermark, time window), and host/peer attestations.

- **Verification Pipeline:** rules + ML heuristics; randomized audits; dispute/appeal queue with **bounded SLAs**.
- **HII Scoring:** configurable weights and decay (α , β , δ , λ), dampening for repetitive tasks, and caps per period.

D. Incentive & Treasury Layer

- **Rewards Contract:** accrual ledger; **claim-time issuance burn** (DAO-tunable bps); budget ceilings per epoch.
- **Treasury & Buyback Module:** SOP-governed; TWAP-guarded buybacks executed only from net realized profit; circuit breakers + oracle freshness checks; public logs. (See Annex E: Oracle Providers & Max-Age table.)
- **Staking/Locking:** optional modules to reduce velocity and align longer-term participation.

E. Commerce Layer (Peace Store)

- **Merchant Registry:** tiered onboarding with risk scoring and compliance flags.
- **Order Router:** discount logic linked to HII tiers; anti-wash filters; dispute resolution primitives.
- **Fee Splitter:** programmable splits among treasury, hosts/NGOs, and ecosystem funds; all distributions logged.

F. Governance Layer (DAO)

- **Proposal → Vote → Timelock → Execute** flow with role isolation for critical ops (treasury moves, registry changes).
- **Safety Rails:** vote caps per ID, HII-weighting, class-based quorums; explicit delay windows; emergency-stop playbooks.
- **Config Registry (On-Chain):** single source of truth for parameters (reward bps, buyback windows, oracle max-age, HII weights, discount elasticity).

G. Data & Observability (Indexers & KPIs)

- **Indexer:** assembles event streams into analytics tables (missions, rewards, buybacks, GMV with anti-wash filters).
- **KPI Service (Public):** exposes dashboards and signed monthly reports; hashes anchored on-chain for tamper-evidence.
- **Forensics Hooks:** export endpoints (CSV/JSON/Parquet) for third-party analytics.

4.5 Component Deep-Dive

4.5.1 Identity: Peace Passport cNFT

- **Minting & Binding:** One cNFT per verified human; transfer disabled; revocation only via governance-approved SOP.
- **Metadata Strategy:** Minimal PII; pointers to attestations and jurisdiction flags; proof hashes (not raw data) for privacy.
- **Interfaces:** `Passport.mint(address, attestHash)`, `Passport.revoke(id, reason)`, `Passport.flags(id)`.

Rationale: Sticky identity is essential for fair rewards and governance caps; non-transferability prevents reputation arbitrage.

4.5.2 Missions & Verification

- **Mission Template Registry:** `MissionType` definitions with required signals, valid ranges (geo-fences, time windows), and reward curves.
- **Submission Flow:** client → signed payload → gateway → verification pipeline → on-chain commit (hash) → reward accrual.
- **Randomized Audits:** adjustable sampling rate; **host attestations** and **peer checks**; appeals with tight SLAs; penalties escalate for repeated abuse.
- **HII Mechanics:** Base points × mission multiplier × (1 – repetition decay) × (veracity factor). Seasonal caps prevent farming.

Why it works: Fraud becomes **costly** (multi-signal requirements + audit risk) while legitimate users enjoy **fast, predictable** confirmations.

4.5.3 Rewards, Burn, and Budget Discipline

- **Accrual vs. Claim:** Rewards accrue off-chain in the indexer for speed, then **claim** triggers an on-chain mint **minus burn bps** set in the Config Registry.
- **Budget Ceiling:** Epoch-level reward cap; graceful degradation (queue-overflow delay, not silent mint overrun).
- **Treasury Inflow/Outflow:** Inflows from fees and partnerships; outflows to buybacks are guarded (TWAP windows, liquidity checks, oracle freshness, coverage threshold). (See Annex E: Oracle Providers & Max-Age table.)
- **Transparency:** Every buyback emits `BuybackExecuted(id, baseAsset, notional, twapWindow, proofHash)`; monthly consolidated report hash anchored on-chain.

4.5.4 Peace Store & Anti-Wash Commerce

- **HII-Tiered Discounts:** Elasticity tuned in registry; per-category caps; cooldowns on serial redemptions.
- **Anti-Wash Filters:** velocity thresholds, same-entity loops, abnormal redemption patterns; automated holds pending review.
- **Merchant Lifecycle:** Onboarding → probation (enhanced monitoring) → graduated tiers; slashing of fee share for non-compliance.

Design goal: Convert verified impact into **spendable utility** while protecting merchants from coupon arbitrage.

4.5.5 Governance & Config Registry

- **Proposal Classes:** “Operational” (low risk), “Economic” (reward/buyback tuning), “Critical” (treasury moves, emergency flags). Each class has distinct quorum and timelock.
- **Voting Mechanics:** Cap per identity; **HII-weight option** to elevate high-contributor voices; quadratic component (optional) to soften capital dominance.
- **Registry Discipline:** Any parameter change emits `ConfigUpdated(key, old, new, proposer, eta)`; **no** behind-the-scenes toggles.

Outcome: A measured pace of change that is visible, deliberate, and revertible.

4.5.6 Data Plane & Public KPIs

- **Canonical Tables:** `missions_verified`, `rewards_claimed`, `store_gmv_clean`, `buybacks_executed`, `governance_votes`, `circulating_supply`.
- **Derived Metrics:**
 - **Mission Verification Rate (MVR)** = verified / submitted.
 - **De-Washed GMV** = GMV – (flagged/wash transactions).
 - **Token Velocity Decay Rate (TVDR)** = $\Delta(1/\text{velocity})$ over rolling windows.
 - **Ecosystem Resilience Score (ERS)** = composite of coverage months, GMV stability, and stake/lock ratios.
- **Publishing:** Live dashboards + monthly PDF/CSV dumps with content hashes anchored on-chain.

4.6 On-Chain & Off-Chain Research Plan

On-Chain (Index & Analyze):

- Track wallet clusters for top participants (hosts, merchants, heavy claimants).
- Build funnel analytics from `missions_submitted` → `missions_verified` → `rewards_claimed` → `store_spend`.
- Monitor governance power concentration (Herfindahl index) **after** vote caps and HII-weighting.
- Publish **TVDR**, **ERS**, and **Coverage Months** in a public endpoint.

Off-Chain (Corroborate & Calibrate):

- Host audits (randomized) and merchant satisfaction surveys; correlate disputes vs. merchant tier.

- Policy watchlist by jurisdiction (KYC thresholds, sanctions lists) → parameter toggles via registry proposals.
- Community listening (forums, social, support desk) with tagged themes that map to product sprints.

Embed code (template):

```
<iframe src="https://analytics.peacesoldiers.org/embed/dashboard?id=impact-funnel%22
width="100%" height="480" frameborder="0"></iframe>
```

4.7 Security Model & Threat Mitigations

Threat	Vector	Primary Controls	Residual Risk
Sybil / Multi-accounting	Recycled identities	Soulbound passport, duplicate-signal checks, device heuristics, KYC tiers	Low–Med
Proof forgery	GPS spoofing, fabricated media	Multi-signal rules, watermarks, host attestations, randomized audits, penalties	Low–Med
Wash commerce	Self-dealing loops	Merchant risk scoring, velocity caps, pattern detection, dispute engine	Medium
Governance capture	Vote concentration, flashIdentity caps, HII weight, class-based orgs	quorums, timelocks	Low–Med
Oracle/manipulation	Stale prices, thin liquidity	Oracle freshness checks, TWAP windows, circuit breakers	Low
Key/ops risk	Misconfigured params	Registry-only changes, 4-eyes review, SOP drills, immutable logs	Low

4.8 Creative Framework — “The Parametric City”

Imagine the protocol as a **city**:

- **Streets (Access)**: MiniApp routes people quickly and safely.
- **Passports (Identity)**: One person, one credential; visas (attestations) unlock districts.
- **Zoning (Proof & HII)**: District rules (signals, caps, audits) keep neighborhoods healthy.
- **Utilities (Incentives)**: Budgeted power/water (rewards issuance) with meters (burn bps) and green policies (buybacks on net profit).
- **Commercial districts (Store)**: Merchants compete fairly under anti-wash bylaws.
- **City council (DAO)**: Ordinances pass through hearings (timelocks), supermajorities for critical changes, and public records.
When traffic grows, you **add lanes** (scale access), **tune lights** (update parameters), and **enforce codes** (audits). The city thrives not by slogans, but by **good zoning and reliable utilities**.

4.9 Rollout Plan

Phase 0 — Dry Run (Internal):

- Spin up indexer & KPI service; simulate mission types; run audit drills; seed test merchants.

Phase 1 — Closed Beta:

- Mint initial Peace Passports; cap mission volume; enable appeals; publish first KPI dashboard.

Phase 2 — Open Beta:

- Expand merchants; enable buyback module in **monitor** mode (no execution) until coverage ≥ 12 months; stress-test Store anti-wash.

Phase 3 — Global Launch:

- Turn on buyback execution within SOP limits; broaden DAO classes; finalize public data exports.

Phase 4 — Progressive Decentralization:

- Hand off upgrade keys; enforce stricter proposal classes; lock emergency powers behind supermajority.

4.10 Quality Gate

- **Fact pattern coherence:** Each control surface ties directly to a Chapter 3 problem statement.
- **Engineering sanity:** Parameters live in an on-chain registry; critical actions gated by timelocks/SOPs; telemetry public.
- **Contrarian review (brief):**
 - *Challenge:* Buybacks can be performative if not backed by real profits.
Response: Execution is **profit-conditioned** with coverage guards and TWAP; no “optics-only” intervention.
 - *Challenge:* MiniApp UX ≠ security.
Response: All security-critical actions (rewards, registry changes) are on-chain and observable; gateway is stateless and tightly scoped.
 - *Challenge:* Identity can creep into privacy debt.
Response: Minimal PII, off-chain erasure pathways, and on-chain hash anchoring strike the balance.

Chapter 5 — Technical Specifications

5.1 Executive Summary

This chapter translates Peace Soldiers into **concrete, testable engineering artifacts** on Solana. The system is a set of well-scoped programs (smart contracts), a hardened off-chain perimeter (gateways, verifiers, indexers), and a **single on-chain Config Registry** that parameterizes behavior without redeploys. Everything critical is (a) observable via events, (b) governed through timelocked proposals, and (c) bounded by safeguards (caps, circuit breakers, TWAP windows).

5.2 Key Findings

1. **Programs are small and composable.** Each contract has a narrow responsibility—Identity, Missions, Rewards, Treasury, Store, Governance, Registry—reducing blast radius and easing audits.
2. **Parameters live on-chain.** No “hidden flags.” All tunables (burn bps, buyback windows, HII weights, discount bands, oracle max-age) reside in a versioned **Config Registry PDA**.
3. **Credible discipline is enforced in code.** Rewards mint only on claim with burn bps; buybacks execute only from realized profit and under TWAP/coverage guards; governance changes wait behind timelocks.
4. **Observability is first-class.** Emitted events form a canonical log; an indexer materializes dashboards and exports (Parquet/CSV/JSON) for third-party replication.
5. **Upgrades are gated.** Program upgrades require proposal class “Critical,” supermajority quorum, and time delay; emergency procedures are codified via SOP transactions, not ad hoc ops.

5.3 Methodology

- Started with the **problem-to-surface mapping** (Ch.3) → designed **one program per surface**.

- Applied **threat modeling** (Sybil, forgery, wash, governance capture, oracle drift) to each surface → added guardrails.
- Chose **event-first design**: every state transition emits a normalized event.
- Set **SLAs and compute budgets** to keep UX fast while preventing spam.

5.4 Program Suite & Responsibilities

5.4.1 Identity Program (**passport**)

- **Purpose**: Non-transferable Peace Passport cNFT issuance and lifecycle.
- **Accounts**:
 - **PassportMint** (SPL mint; **transfer_hook** denies outbound transfers)
 - **PassportPDA(user)** (owner state: status, jurisdiction, attest hash pointers)
- **Key Instructions**:
 - **issue(user, attest_hash)** — mints cNFT if **!has_passport(user)** and pre-checks pass.
 - **revoke(user, reason_code)** — governance-gated.
 - **set_flag(user, flag, value)** — sets compliance and reputation flags.
- **Guards**: one-per-human policy (device/duplicate heuristics enforced off-chain; on-chain enforces one PDA per wallet); revocations require proposal class “Operational” with timelock.
- **Events**: **PassportIssued**, **PassportRevoked**, **PassportFlagUpdated**.

5.4.2 Missions Program (**missions**)

- **Purpose**: Register mission templates; accept submissions; anchor proof hashes; hand off to verification pipeline.
- **Accounts**:

- `MissionTypePDA(id)` — E/V/A types, required signals, base score, caps.
- `MissionSubmissionPDA(id,user)` — status, proof hash, verifier score, timestamps.
- **Key Instructions:**
 - `register_type(params)` — governance-gated; emits `MissionTypeRegistered`.
 - `submit(user, type_id, proof_hash)` — writes immutable hash + timestamps.
 - `attest(submission_id, host_sig, score)` — host/peer attestations (multisig optional).
 - `finalize(submission_id, verifier_hash, verdict)` — closes loop; sets `eligible_reward_units`.
- **Guards:** per-period submission caps; duplicate hash detection; time/geo windows enforced off-chain but signed into the payload included in the hash.
- **Events:** `MissionSubmitted`, `MissionAttested`, `MissionFinalized`.

5.4.3 Rewards Program (**rewards**)

- **Purpose:** Accrue and mint rewards; apply claim-time burn; enforce budget ceilings.
- **Accounts:**
 - `RewardsPDA(user, epoch)` — accrued units, status.
 - `BudgetPDA(epoch)` — ceiling, consumed.
- **Key Instructions:**
 - `accrue(user, units)` — only callable by `missions.finalize`; increases user's accrual and epoch consumption.
 - `claim(user)` — mints SPL `$PEACE` to user minus `burn_bps` from `ConfigRegistry`; emits detailed receipt.

- **Guards:**
 - Fails if epoch ceiling exceeded;
 - `burn_bps` and `max_claim_per_epoch` read from registry;
 - `CircuitBreaker` halts claims if registry toggled (governance only).
- **Events:** `RewardAccrued`, `RewardClaimed{gross, burn_bps, net}`.

5.4.4 Treasury & Buyback Program (`treasury`)

- **Purpose:** Manage treasury assets, execute **TWAP-guarded** buybacks, log reports.
- **Accounts:**
 - `TreasuryVault` (multi-sig / DAO owned)
 - `BuybackPDA(window_id)` — parameters: `twap_window`, `oracle_max_age`, `coverage_threshold`, `max_notional_per_window`.
- **Key Instructions:**
 - `queue_buyback(window_id, notional)` — schedules within TWAP window; checks coverage/profit flags provided by accounting oracle.
 - `execute_buyback(window_id)` — performs DCA across the window; burns received tokens immediately.
 - `file_report(month, ipfs_hash)` — monthly snapshot hash anchor.
- **Guards:**
 - Requires **net realized profit** flag = true;
 - Enforces `coverage_threshold` months of reward runway;
 - Validates oracle freshness $\leq$ `oracle_max_age`; (See Annex E: Oracle Providers & Max-Age table.)

- Global `EmergencyStop` switch in registry.
- **Events:** `BuybackQueued`, `BuybackExecuted`, `TreasuryReportFiled`.

5.4.5 Store Program (**store**)

- **Purpose:** Merchant registry, order routing, HHI-tiered discounts, anti-wash guards.
- **Accounts:**
 - `MerchantPDA(id)` — KYC tier, risk score, fee split.
 - `OrderPDA(id, user, merchant)` — order meta; discount applied; flags.
- **Key Instructions:**
 - `register_merchant(params)` — role-gated; emits `MerchantRegistered`.
 - `place_order(user, merchant, amount, hhi_tier)` — applies discount policy; emits `OrderPlaced`.
 - `flag_order(id, reason)` — holds settlement pending review.
- **Guards:** velocity thresholds per user/merchant; loop detection (user ↔ merchant graph anomalies); cooldowns; per-category discount caps from registry.
- **Events:** `MerchantRegistered`, `OrderPlaced`, `OrderFlagged`, `OrderSettled`.

5.4.6 Governance Program (**dao**)

- **Purpose:** Proposals, voting, timelocks, execution routing.
- **Accounts:**
 - `ProposalPDA(id)` — class, targets, calldata, ETA, state.
 - `BallotPDA(proposal, user)` — vote power snapshot.

- **Key Instructions:**
 - `propose(class, targets, calldata)` — creates proposal; class determines quorum and delay.
 - `cast_vote(user, weight)` — supports HII-weighted or pure-token modes per registry.
 - `queue(id) / execute(id)` — enforces timelock; dispatches to targets.
- **Guards:** cap per identity; minimum participation for Critical class; veto-less design—use supermajority quorum instead.
- **Events:** `ProposalCreated`, `VoteCast`, `ProposalQueued`, `ProposalExecuted`, `ConfigUpdated`.

5.4.7 Config Registry Program (`config`)

- **Purpose:** Single source of truth for parameters; append-only history with diffs.
- **Accounts:**
 - `ConfigRoot` — current pointers;
 - `ConfigVersionPDA(version)` — key/values, author, proposal id, ETA, timestamp.
- **Key Instructions:** `set(key, value)`, `promote(version)`, `freeze()` for emergency.
- **Events:** `ConfigSet`, `ConfigPromoted`, `ConfigFrozen`.

5.5 Off-Chain Perimeter & Data Plane

5.5.1 API Gateway

- Stateless; verifies MiniApp signed payloads; rate-limits per passport; forwards only to allow-listed routes.

- **SLA:** p95 < 200 ms; autoscaled; WAF in front.

5.5.2 Verification Pipeline

- **Signal adapters:** GPS consistency, device attestation, watermark/time drift, host signature checks.
- **Rules engine:** per-mission template thresholds; adaptive sampling for audits.
- **Disputes:** bounded SLA ($\leq 72h$) with canonical outcomes posted on-chain as hashes.

5.5.3 Indexer & Analytics

- **Sources:** program logs & accounts; de-duplication; late-arriving correction merges.
- **Tables:** `missions_verified`, `rewards_claimed`, `store_gmv_clean`, `buybacks_executed`, `governance_votes`, `circulating_supply`.
- **Derived metrics:** MVR, TVDR, ERS, Coverage Months, Gini/Herfindahl of vote power.
- **Exports:** hourly Parquet to object storage; signed monthly digest; public dashboards.

5.6 Data Schemas (selected)

Event: `RewardClaimed`

```
{
  "event": "RewardClaimed",
  "user": <pubkey>,
  "epoch": <u32>,
  "gross_units": <u64>,
  "burn_bps": <u16>,
  "net_minted": <u64>,
  "tx": <sig>,
  "ts": <unix_ms>,
  "config_version": <u32>
}
```

Account: **MissionSubmissionPDA**

```
{
  "id": <u128>,
  "user": <pubkey>,
  "type_id": <u32>,
  "status": <enum: Submitted|Attested|Finalized|Rejected>,
  "proof_hash": <bytes32>,
  "verifier_hash": <bytes32>,
  "eligible_units": <u64>,
  "created_at": <unix_ms>,
  "finalized_at": <unix_ms|null>
}
```

Account: **ConfigVersionPDA (partial)**

```
{
  "version": <u32>,
  "params": [
    {"key": "burn_bps", "val": "75"},
    {"key": "buyback_twap_window_s", "val": "7200"},
    {"key": "coverage_threshold_months", "val": "12"},
    {"key": "oracle_max_age_s", "val": "120"},
    {"key": "discount_cap_pct", "val": "30"}
  ],
  "author": <pubkey>,
  "proposal_id": <u64>,
  "eta": <unix_ms>,
  "ts": <unix_ms>
}
```

5.7 Security, Gas/Compute, and Rate Limits

- **Access Control:** CPI calls restricted; instruction ACLs checked against `dao.execute` or program-owned PDAs.
- **Key Management:** Multisig for vaults; upgrade authority behind Critical proposals; rotateable signers with on-chain announcements.
- **Compute Budgets:** Target $\leq 100k$ CU per hot path (submissions, claims, orders); preflight checks avoid wasted CU.

- **Fees/Priority:** Client chooses priority fees for congested windows; guards ensure economics remain viable (max fee bps in registry).
- **Rate Limits:** Per-passport cooldowns; per-merchant velocity; API gateway token bucket per IP and passport.
- **Oracles:** Medianized price feeds; freshness enforced by `oracle_max_age_s`; fallback halts buybacks.
- **Circuit Breakers:** Global halts for claims, buybacks, store orders; each requires governance action to release.
- **Logging:** All halts/reenables emit `CircuitBreakerToggled`.

5.8 Testing & Audit Plan

- **Unit Tests:** 90%+ branch coverage for each program; golden tests for registry diffs and timelock edge cases.
- **Property Tests:** Invariants—epoch ceilings never exceeded; claim always $\geq$ burn; buyback only with profit & coverage.
- **Fuzzing:** Inputs for submissions, discount elasticity, TWAP sequencing.
- **Testnets:** Devnet (rapid), Testnet (soak); canary deploy with read-only dashboards.
- **External Audit:** Two independent firms; scope includes governance replay, oracle liveness, and registry upgrade path.
- **Red Team Drill:** Simulated wash and governance-capture scenarios; runbooks validated.
- **SOP Drills:** Quarterly emergency halts and resume sequences; published outcomes.

5.9 Operational SLAs & Telemetry

- **Gateway p95:** < 200 ms; error rate < 0.5%.
- **Indexer Lag:** < 60 s behind chain head.

- **Verification SLA:** auto-verdict < 5 s; manual appeals ≤ 72 h.
- **Dashboard Freshness:** near-real-time tiles; monthly signed digest (hash on-chain).
- **Governance Timelocks:** class-based (e.g., Operational ≥ 12h; Economic ≥ 48h; Critical ≥ 96h).
- **Incident Transparency:** P0 post-mortem within 72 h; on-chain hash of report.

5.10 Creative Analogy — “Air Traffic Control for Impact”

Think of Peace Soldiers as **air traffic control**:

- **Flight plans** = mission templates.
- **Transponders** = passports broadcasting identity signals.
- **Radar & towers** = verification pipeline and indexer visibility.
- **Runway slots** = budget ceilings and rate limits.
- **Fuel policy** = reward issuance with burn bps.
- **Noise abatement** = TWAP-guarded buybacks and circuit breakers.
Every landing (claim) is scheduled, logged, and audited. The skies stay safe not by hoping for good pilots, but by **procedures and instrumentation**.

5.11 Rollout Checklist (Engineer’s Cut)

- Deploy **config** (empty), lock upgrade keys in multisig.
- Deploy **passport**; dry-run issuance; negative tests for transfer hooks.
- Deploy **missions**; register 3 baseline mission types; verify event flow.
- Deploy **rewards**; set **burn_bps**, **epoch_ceiling**; simulate accrual/claim.
- Deploy **store**; onboard 5 merchants in probation tier; exercise flags.
- Deploy **treasury**; queue dummy buyback with oracle; verify guards.

- Deploy `dao`; pass first “Operational” and “Economic” proposals; promote `config` v1.
- Wire indexer; publish first public dashboard; anchor monthly digest hash.
- Execute SOP drills (EmergencyStop, parameter rollback, oracle stale).
- Freeze non-essential toggles until post-beta review.

Chapter 6 — Tokenomics

6.1 Executive Summary

The \$PEACE economy is engineered for **discipline, utility, and transparency**. Supply is **fixed**; issuance occurs only at **claim**; every claim applies a **programmatically burn**; buybacks are **conditioned on realized net profit** and executed via **TWAP windows** under strict coverage guards. Utility spans: **governance**, **discounted commerce** (Peace Store), **staking/locking** (velocity control), and **reputation coupling** (HII-weight options). Reporting is continuous: circulating supply, reward coverage months, buyback logs, and a public KPI suite. The result is a token that's hard to print, easy to audit, and actually **used**.

6.2 Key Findings

1. **Fixed-supply discipline with real sinks:** Claim-time burn + profit-only buybacks ensure that net supply trends are driven by verified activity and productive revenue, not discretionary minting.
2. **Separation of concerns:** Rewards, treasury, and governance are distinct programs with on-chain parameters; no “hidden toggles.”
3. **Coverage-first economics:** Incentives scale with **budget ceilings** and **coverage months**, avoiding reflexive emissions.
4. **Utility beats speculation:** HII-tiered store discounts, governance access, and partner integrations create non-speculative demand.
5. **Transparency by default:** Every material action leaves a machine-readable trail—burn bps, buyback windows, and emissions are all observable.

6.3 Methodology

- **Top-down design:** Start with end-state constraints (fixed supply; solvency under stress), then back into parameters (burn bps, budget ceilings, buyback cadence).
- **Bottom-up stress:** Model edge cases—adoption spikes, mission farming attempts, liquidity droughts—and size ceilings and circuit breakers.

- **Metric-first instrumentation:** Define and publish TVDR, Coverage Months, ERS, de-washed GMV, and governance concentration indices from day one.

6.4 Deep-Dive Analysis

6.4.1 Supply, Issuance, and Burn

- **Total Supply:** Fixed at genesis; no mint authority outside governed claims.
- **Accrual vs. Claim:** Users **accrue** rewards off-chain after mission finalization. **Only at claim** does on-chain minting occur, immediately applying **burn_bps** (e.g., 0.75%–3.00%, DAO-tunable).
- **Rationale:** Claim-time burn ties net issuance to **user-initiated conversion**—if users don't claim, supply doesn't expand; if they do, expansion is dampened.
- **Parameter Discipline:** **burn_bps** changes via proposal, timelock, and on-chain diff.

6.4.2 Rewards Budget & Coverage

- **Epoch Ceiling:** Each epoch has a **hard cap** of reward units; accruals beyond the cap queue, they don't overrun.
- **Coverage Months:** Treasury plus forecasted inflows must cover $\geq N$ months of rewards at current run-rate before enabling any buybacks or APR raises.
- **Adaptive Levers:** DAO can adjust mission multipliers, caps, and burn bps to maintain coverage.

6.4.3 Buyback & Burn — Profit-Conditioned

- **Funding Source:** Only **realized net profit** (after operating costs) may fund buybacks.
- **Execution:** TWAP windows with oracle freshness checks and per-window notional caps; tokens purchased are burned immediately. (See Annex E: Oracle Providers & Max-Age table.)
- **Circuit Breakers:** If liquidity is thin or oracle is stale, buyback halts automatically.
- **Why it matters:** This avoids treasury depletion theater and aligns token support with productive activity.

6.4.4 Token Utility (Demand Surfaces)

- **Governance:** Proposal/vote participation, with optional **HII-weight** and identity vote caps.
- **Peace Store:** HII-tiered discounts and partner offers create everyday demand; anti-wash filters protect merchant economics.
- **Staking/Locking (optional):** Velocity control and access tiers; parameters in the Config Registry.
- **Partner Integrations:** Access gates and fee rebates for ecosystem services.

6.4.5 Distribution & Vesting (Principles)

- **Team/Partner Locks:** Cliff + linear vesting, hard-coded timelocks, public schedules, and a circulating-supply dashboard.
- **Community/Rewards:** Majority earmarked for long-term incentives; subject to epoch ceilings and burn bps.
- **Treasury:** Multi-sig/DAO controlled; explicit policy hierarchy (coverage > operations > buyback).

6.4.6 Emission & Velocity Dynamics

- **TVDR (Token Velocity Decay Rate):** Tracks the **rate at which velocity slows** as staking/locking and real utility rise.
- **Sink Elasticity:** HII-tiered Store discounts and periodic partner events shift marginal holders to spend/use rather than churn.
- **Governance Stickiness:** Reputation-coupled voting power (with caps) disincentivizes dumping by active contributors.

6.4.7 Economic Stress Tests

Triggers: (a) Coverage Months < threshold; (b) Clean GMV/Subsidy below floor; (c) TVDR negative drift for N windows. Actions are recorded on-chain via ConfigUpdated and RewardPolicyAdjusted events.

Policy: The protocol does not market or promise APR. Rewards are mission-linked and bounded by epoch ceilings. When behavioral or macro signals suggest speculative churn—e.g., TVDR

deterioration or Clean GMV ratio below target—reward multipliers automatically downshift for the next epoch by a DAO-tunable percentage to preserve coverage months and prioritize utility-driven demand.

6.4.8 No-APR Marketing & Dynamic Throttling

- **Adoption Spike:** If verified missions surge 10× in a week, epoch ceilings and repetition-decay keep accruals in check; users may queue to claim across epochs.
- **Liquidity Winter:** Buybacks remain **off** without realized profit; DAO can temporarily raise burn bps within bounds, preserving runway.
- **Merchant Wash Attempts:** Anti-wash filters remove circular flows from GMV; Store discount elasticity tightens until anomalies subside.
- **Governance Attack:** Identity caps, HII-weight, and class-based quorums throttle capture attempts; timelock surfaces attacks before execution.

6.5 Risk Matrix

Risk	Description	Likelihood	Impact	Mitigation
Runaway issuance	Excess claims vs. budget	Medium	High	Epoch ceilings; claim-time burn; dynamic mission multipliers
Treasury depletion	Buybacks during low/negative P&L	Low	High	Profit-only rule; coverage threshold; TWAP + oracle freshness
Speculative churn	Velocity spikes, weak utility	Medium	Medium	HII-linked discounts; locking tiers; steady Store campaigns
Governance capture	Vote concentration	Low–Med	High	Cap per ID; HII-weight; class quorums; timelocks
Data opacity	Unverifiable numbers	Low	High	Public KPIs; monthly signed digests; on-chain hashes

6.6 Creative Scenarios

1. “Thermostat Economy”

Imagine \$PEACE as a **climate-controlled room**. **Burn bps** is your thermostat; **epoch ceilings** are insulated walls; **buybacks** are sunlight—welcome, but weather-dependent. The system stays comfortable by tuning the thermostat and refusing to pretend there’s sun when there isn’t.

2. “Tide Pool Liquidity”

During low tide (bear markets), organisms survive by using less water. Likewise, buybacks pause without real profit; discounts and partnerships deepen local utility so the pool remains alive until the tide returns.

3. “Two-Key Safe”

Rewards and treasury require different keys: users **choose to claim** (key one) and the DAO **chooses to buy back** only with profit (key two). Both must turn; either alone can’t drain the safe.

6.7 Metrics & Telemetry

- **Circulating Supply** (live + 30-day delta)
- **Reward Coverage Months** (runway at current burn bps and accrual rate)
- **TVDR** (rolling 30/90 days)
- **ERS (Ecosystem Resilience Score)** — composite of coverage, GMV stability (de-washed), and stake/lock ratios
- **De-Washed GMV** (Store) and **Discount Elasticity** vs. HII tiers
- **Buyback Log** (TWAP windows, notional, burn confirmation)
- **Governance Concentration** (Herfindahl/Gini after identity caps & HII-weight)

Data Hygiene: All series emitted as events → indexed → exposed via public dashboards and monthly signed digests.

6.8 Quality Gate

- **Fact pattern:** Fixed supply, claim-time burn, profit-conditioned buybacks, epoch ceilings, and coverage guards are encoded as **on-chain parameters** with timelocks.
- **Coherence check:** Incentive surfaces (governance, Store, staking/locking) map directly to non-speculative demand.
- **Contrarian review (brief):**
 - *Objection:* Burns can be cosmetic if issuance is huge. **Answer:** Hard **epoch ceilings** constrain issuance; burns cut net supply only on user-requested mints.
 - *Objection:* Buybacks may chase price. **Answer:** **Profit-only + TWAP + coverage threshold** remove reflexivity; no “optics-only” interventions.
 - *Objection:* Discount demand is transient. **Answer:** Discounts are **Hill-tiered** and budget-capped; repeat-merchant rate and de-washed GMV are monitored, tightening elasticity if needed.

Chapter 7 — Governance

7.1 Executive Summary

Governance in Peace Soldiers is built to be **measured, tamper-evident, and capture-resistant**. It balances broad participation with disciplined change control. The design centers on a **classed proposal system** (Operational, Economic, Critical), **identity-aware voting** (one-person caps with optional HII-weighting and quadratic dampening), **timelocks** that create review windows, and **SOP-codified** emergency responses. Parameters live in a **single on-chain Config Registry** so that any change is **visible, revertible, and attributable**. Treasury interventions (e.g., buybacks) remain **profit-conditioned** and bound by coverage guards. The result is a system that moves at a deliberate pace—fast enough to operate, slow enough to avoid “overnight governance risk.”

7.2 Key Findings

1. **Separation of powers:** Governance directs policy; programs enforce mechanics. Registry updates, not “ad hoc ops,” drive behavior.
2. **Identity before influence:** Vote caps per human (Peace Passport) and optional **HII-weighting** prevent plutocratic dominance and reward real contribution.
3. **Change control that scales:** Three proposal classes with escalating quorum and timelocks align decision tempo with risk.
4. **Defense-in-depth:** Timelocks, pause/circuit breakers, and SOPs bound blast radius; buybacks are gated by **profit + coverage**—not sentiment.
5. **Audit-ready transparency:** Every material action emits events, publishes diffs, and logs a signed digest; governance health is metrified (Herfindahl, turnout, timelock utilization).

7.3 Methodology

- **Design goals extracted** from Chapters 3–6 (prevent capture, keep incentives solvent, ensure reversibility), mapped to governance primitives.

- **Threat modeling** for vote concentration, flash mobs, proposal stuffing, treasury misuse, and oracle manipulation.
- **Control alignment** to ensure each risk has at least two orthogonal mitigations (e.g., identity caps + timelocks).

7.4 Governance Topology

7.4.1 Participants & Roles

- **Citizens (Passport holders):** May create proposals (with threshold), vote, and delegate.
- **Stewards (elected):** Curate proposal quality, manage disclosures, coordinate audits; no unilateral power.
- **Council (multisig executor):** Signs only transactions emitted by `dao.execute` after timelocks; cannot bypass Registry.
- **Auditors (external/DAO-mandated):** Conduct periodic reviews; publish findings anchored on-chain.
- **Program Maintainers:** Prepare upgrades gated by Critical proposals and timelocks; publish diffs and migration plans.

7.4.2 Proposal Classes

- **Operational:** UI text, non-financial configs, audit scheduling. Lower quorum; **TL ≥ 12h**.
- **Economic:** Reward multipliers, burn bps bands, discount elasticity, sampling rates. Medium quorum; **TL ≥ 48h**.
- **Critical:** Treasury movements, buyback policy changes, program upgrades, EmergencyStop toggles. Supermajority quorum; **TL ≥ 96h**.

All classes produce a machine-readable diff against the Config Registry; Critical actions require an accompanying rollback plan.

7.4.3 Voting Mechanics

- **Eligibility:** One voting identity per Peace Passport (soulbound).

- **Caps:** Max effective vote per ID (e.g., $\leq 0.1\%$ of total voting power snapshot).
- **Weighting modes:**
 - **Identity-only** (1 person = 1 capped vote);
 - **Hybrid (HII-weighted)** where verified contributors receive modest weight boosts;
 - **Quadratic overlay** (optional) to soften large token blocks.
- **Delegation:** Explicit, revocable, snapshot-based; caps **apply after** delegation aggregation.
- **Quorums:** Class-specific; inactivity cannot pass changes—abstain $\neq$ yes.

7.4.4 Timelocks & Execution

- **Queue:** After a successful vote, proposal is queued with $\text{ETA} \geq \text{class timelock}$.
- **Transparency window:** Off-chain reviews, public comments, auditor spot checks.
- **Execute:** `dao.execute` routes calldata to targets; Council signs transactions derived from proposal hash.
- **Immutability sheath:** Registry versions remain accessible; any execution emits `(proposal_id, version_before, version_after)`.

7.5 What Governance Controls (Scope)

1. **Config Registry:** Reward bps, epoch ceilings, buyback twap windows, oracle max-age, discount caps, audit sampling rates, HII weights, compliance toggles.
2. **Treasury Policy:** Profit definition, coverage thresholds, permitted venues, per-window notional caps, reporting cadence.
3. **Program Upgrades:** Hash-pinned binaries, migrations, rollbacks.
4. **SOP Activation:** EmergencyStop toggles (claims, buybacks, store orders), incident response escalation paths.

5. **Grants & Partnerships:** Budget ranges, milestone gates, clawback clauses anchored on-chain.

Governance does **not** perform manual transfers outside approved flows; it sets parameters and approves automated execution paths.

7.6 Integrity & Safety Rails

7.6.1 Capture Resistance

- Identity vote cap per human.
- Optional HII weight for civic contributors.
- Quadratic voting overlay for token blocks.
- Classed quorums + elevated supermajority for Critical proposals.
- **Cooling-off:** Same parameter cannot be flipped back within a minimal dwell time (e.g., 7 days) unless Emergency SOP.

7.6.2 Proposal Hygiene

- **Disclosure pack** mandatory: summary, risk analysis, diffs, metrics impact (coverage months, TVDR), and rollback.
- **Conflict-of-interest attestation** for proposers and major delegates.
- **Single-subject rule** (no proposal stuffing).

7.6.3 Treasury Guardrails

- **Profit-conditioned buybacks** only; enforced by accounting oracle + coverage months check.
- TWAP + oracle freshness to avoid price manipulation. (See Annex E: Oracle Providers & Max-Age table.)
- **Per-window notional caps** to limit slippage and signal control.
- **Monthly digest hash** anchored on-chain with breakdown.

7.6.4 Emergency Response (SOP)

- Triggers: oracle stale, exploit indicators, governance process corruption, regulatory injunctions.
- Steps: pause specific surfaces → publish incident note & ETA → forensic review → staged resume under Council supervision → post-mortem anchored on-chain.

7.7 Governance Data Plane

Core Metrics (public dashboards):

- **Turnout by class** (Operational/Economic/Critical).
- **Effective voting power distribution** (Herfindahl, Gini) after caps and HII.
- **Proposal latency** (creation → queue → execute).
- **Registry churn** (# of parameter changes / 30 days).
- **Incident log** (halts, resumes, SOP drills).
- **Economic health overlays** (Coverage Months, TVDR, de-washed GMV) next to each Economic/Critical change.

Publishing cadence: real-time tiles + **monthly signed digest**; all reports include content hashes and query fingerprints.

7.8 Governance Processes (Playbooks)

7.8.1 Parameter Change (Economic)

1. Draft with impact analysis → 2) Steward review (format only) → 3) Submit → 4) Vote → 5) Queue (≥48h) → 6) Execute → 7) Monitor KPI deltas for 30 days → 8) Auto-review checkpoint.

7.8.2 Treasury Buyback Adjustment (Critical)

- Preconditions: **profit true**, **coverage ≥ threshold**, **oracle fresh**.

- Proposal includes: TWAP window, notional cap, venues, and rollback.
- Post-execute: publish execution receipts and updated coverage.

7.8.3 EmergencyStop Engage/Release

- Engage: Critical proposal (fast-track path within bounds) → immediate halt → incident note within 24–72h.
- Resume: staged re-enable with reduced limits; post-mortem required.

7.8.4 Grants / NGO Budget Allocation

- Milestone-based disbursement with on-chain milestones; clawback if KPIs missed; public ledger of outcomes (hashes).

7.9 Risk Matrix

Risk	Description	Likelihood	Impact	Mitigations
Vote concentration	Large holder or delegate cartel	Med	High	Identity caps, HII weight, quadratic overlay, classed quorums, timelocks
Flash governance	Rapid, complex changes without scrutiny	Low–Med	High	Timelocks, disclosure packs, single-subject rule, transparency window
Treasury misuse	Buybacks without profit/coverage	Low	High	Profit-conditioned rule, oracle freshness, TWAP, per-window caps
Proposal stuffing	Bundling unrelated changes	Med	Med	Single-subject rule, Steward checks, voter UI highlighting diffs
Process capture	Council or maintainer key abuse	Low	High	<code>dao.execute</code> -only routes, multisig limits, public signer rotations

Regulatory shocks	Sudden KYC/AML shifts	Med	High	Config toggles, jurisdiction flags, timelocks for scoped rollout
Voter apathy	Low turnout, easy capture	Med	Med	Default delegation, participation rewards (non-monetary recognition), reminders

7.10 Creative Framework — “The Three-Chamber Heart”

Picture governance as a **three-chamber heart**:

- **Atrium (Operational)**: High-frequency, low-risk blood flow—keeps daily life steady.
- **Ventricle (Economic)**: Strong, paced contractions—push policy that feeds the economy.
- **Aortic Valve (Critical)**: Slow, high-pressure decisions—once open, must be deliberate and well-timed.
The heart keeps rhythm through **pacemakers** (timelocks), **blood pressure** (quorums), and **oxygenation** (transparent data). Too fast? Arrhythmia (flash governance). Too slow? Ischemia (stagnation). The design aims for **healthy cadence**.

7.11 Progressive Decentralization (Roadmap)

- **Phase A — Bootstrap**: Core team proposes; Council executes; strict caps; frequent audits.
- **Phase B — Shared Control**: Community stewards elected; expanded proposer set; Economic changes migrate fully on-chain.
- **Phase C — Mature DAO**: Critical upgrades require supermajority and external audit sign-off; emergency powers behind stricter thresholds; key rotations on schedule.

7.12 Quality Gate

- **Fact pattern coherence**: Every governance control ties to an explicit risk and emits observable artifacts (events, diffs).

- **Safety proofing:** Dual layers (identity caps + timelocks; profit rule + TWAP) mitigate single-point failures.
- **Contrarian review (brief):**
 - Objection: HII-weighting could bias toward “popular” impact. **Answer:** HII weights are bounded and transparent; proposal classes prevent outsized influence; audits verify metric integrity.
 - Objection: Timelocks slow response. **Answer:** Emergency SOP exists for scoped halts; Critical changes still require review to avoid irreversible errors.
 - Objection: Delegation centralizes power. **Answer:** Caps apply after delegation aggregation; dashboards expose delegate concentration; re-delegation is instant.

Chapter 8 — Human Impact Index (HII) & Missions

8.1 Executive Summary

The Human Impact Index (HII) is the protocol's **economic gravity**—a single, tamper-evident score that translates verified human actions into rights (governance), rewards (token claims), and real-world utility (Peace Store tiers). HII is produced from **multi-signal missions**—Education (E), Volunteering (V), and Activity (A)—each constrained by **templates, proofs, audit sampling, and penalties**. The scoring engine is parameterized on-chain (weights, caps, decay) and observable via events and dashboards. The design goal: **easy to earn honestly, expensive to fake, and precise enough to allocate capital at scale**.

8.2 Key Findings

1. **One score, many consequences.** HII simultaneously gates benefits (discount tiers), influence (HII-weighted voting mode), and access (mission tiers), ensuring impact is not cosmetic.
2. **Multi-signal proofing beats “photo uploads.”** GPS consistency, time-watermarked media, host or peer attestations, and randomized audits make fraud **costly** and **rare**.
3. **Parameter discipline matters.** Four dials—**weights (α, β, δ)**, **repetition decay (λ)**, **caps**, and **audit rate**—control gaming without killing legitimate scale.
4. **Scoring is explainable.** Every HII update references source missions, signals, and rules, enabling dispute resolution and third-party replication.
5. **Privacy-by-construction.** Proof data stays off-chain; **hashes and verdicts** are on-chain with minimal metadata, balancing transparency and compliance.

8.3 Methodology

- **Requirements → Model:** Started from abuse cases (bot farms, media forgery, location spoofing, collusion) and backsolved to a scoring function that stays stable across spikes and jurisdictions.

- **Signal fusion:** Combined deterministic checks (GPS/time/host) with heuristic anomaly detection; tuned thresholds to minimize both false positives (penalizing good users) and false negatives (letting fraud through).
- **A/B fairness:** Calibrated Education vs. Volunteering vs. Activity weights so each category has a path to meaningful HII growth for distinct user profiles.
- **Simulation & red teaming:** Stress-tested repetition decay, daily/weekly caps, and audit sampling against synthetic attack patterns (loop rings, geo-tunnels, device farms).

8.4 HII Design — The Function

Let each verified mission (m) of type (mission type is one of E, V, or A) yield base points (b_t). For user (u) at time (k):

$$\begin{aligned}
 &[\\
 &HII_u(k) = \sum_{m \in \text{mathcal}\{M\}} u(k) \\
 &\underbrace{\alpha_{t(m)}}_{\text{type weight}} b_{t(m)} \\
 &\times \\
 &\underbrace{(1 - \lambda \cdot r_{u,t(m),k})}_{\text{repetition decay}} \\
 &\times \\
 &\underbrace{\phi(m)}_{\text{veracity factor}} \\
 &\times \\
 &\underbrace{\omega_u(k)}_{\text{reputation modifier}} \\
 &]
 \end{aligned}$$

Where:

- ($\alpha_{\{E,V,A\}}$): DAO-tunable type weights (e.g., E=0.8, V=1.0, A=0.6).
- ($r_{u,t,k}$): count of similar missions by (u) in a sliding window; (λ) enforces diminishing returns on spammy repetition.
- ($\phi(m) \in [0, \epsilon, 1]$): veracity factor—1 for verified, (ϵ) (tiny) for weak proofs, 0 for failed audits (plus penalties).
- ($\omega_u(k)$): reputation modifier with soft floors/ceilings based on historical dispute outcomes and host trust.

Caps & Floors:

- **Daily/weekly caps** per type and per user;
- **Tier gates** (e.g., high-tier missions require $HII \geq X$);
- **Decay to zero** for rejected or retracted proofs; **probation** after repeated flags.

8.5 Missions — From Template to Verdict

8.5.1 Mission Templates (E / V / A)

Each template defines: required signals, acceptable ranges, base points, multipliers, and cooldowns.

- **Education (E):** course completion, quizzes, certification checks; signals: time-on-task, proctored checks, content signatures.
- **Volunteering (V):** NGO/host-attested activities with geo/time constraints; signals: host signature, GPS consistency, photo/video watermarking.
- **Activity (A):** social-good micro-actions (clean-ups, community events); signals: crowd presence (Bluetooth/WiFi peers), media entropy, geofence stay time.

Endpoint discipline: Client signs payload (device+timestamp); gateway validates and writes **immutable content hash** on-chain via `missions.submit`.

8.5.2 Verification Pipeline

- **Deterministic checks:** GPS radius, timestamp windows, media watermark and entropy, duplicate-hash guard.
- **Attestations:** Host signatures (weighted by host reputation) and **peer** co-signs (dampened to avoid collusion value).
- **Randomized audits:** Adjustable sampling; escalates with anomalous patterns (e.g., low-entropy photo reuse).
- **Verdict & Scoring:** `finalize(submission_id, verdict, verifier_hash)` → accrual units issued to `rewards` and HII delta logged.

8.5.3 Disputes & Penalties

- **Appeals:** 72h SLA with structured evidence; second-line adjudicators selected from a vetted pool.
- **Penalties:** Zeroing of HII for the mission, **probation flags** on the passport, escalating cool-downs, and slashing of host trust if complicit.

- **Rehabilitation:** Time-based decay of penalties for clean history; avoids permanent lockouts that reduce community goodwill.

8.6 Anti-Fraud: Layers & Costs

Attack Class	Example	Layered Control	Cost to Attacker
GPS spoofing	Fake location apps	Sensor fusion (speed pathing), geofence dwell time, device attestation	High (needs device fleet + path simulation)
Media forgery	Reused stock photos	Perceptual hashing, entropy checks, watermark matching, peer cross-check	Medium-High
Host collusion	Rubber-stamp NGO	Host trust score, cross-mission correlation, random on-site audits	High (risk of slash/ban)
Ring farming	Small group loops	Graph anomaly detection (user ↔ host ↔ merchant), velocity caps	High (needs complex networks)
Device farms	Many low-quality accounts	Soulbound passport, duplicate heuristics, cooldowns per device	High (capital + time)

Principle: Honest behavior should be **low-friction**; successful fraud should be **high-cost and short-lived**.

8.7 HII → Rights, Rewards, and Utility

- **Governance:** Optional **HII-weighted** mode grants modest weight boosts to consistent contributors, bounded by identity vote caps.
- **Rewards:** Some mission tiers require **minimum HII**; repetition decay prevents passive grinding for pure emissions.
- **Peace Store:** **HII tiers** unlock targeted discounts and partner programs; anti-wash logic excludes flagged flows from GMV.

8.8 On-Chain & Off-Chain Research Plan

On-Chain (analytics):

- **MVR (Mission Verification Rate):** verified / submitted, segmented by type, region, and host.
- **HII Distribution:** median, 90th percentile, and Gini to detect concentration or under-rewarding.
- **Repetition Pressure:** share of missions hitting decay zones per user cohort.
- **Fraud Pressure:** rate of flagged → overturned appeals; anomaly incidence per 1,000 missions.
- **Funnel:** submit → verified → accrued → claimed → store spend (de-washed).

Off-Chain (corroboration):

- **Host audits** (random site checks);
- **User surveys** on friction and perceived fairness;
- **Merchant feedback** on coupon abuse and redemption quality;
- **Jurisdiction watchlist** to tune signal requirements (e.g., KYC thresholds, NGO licensing).

Visualization (ready-to-plug):

- **PNG 1:** “HII Curve by Cohort” (P10/P50/P90 vs. weeks since join).
- **PNG 2:** “Fraud Pressure Gauge” (flags, appeals, reversals).

8.9 Fairness & Accessibility

- **Multiple paths to impact:** E/V/A balance lets knowledge workers, caregivers, and field volunteers all advance.

- **Device & bandwidth tolerance:** Offline caching and small media requirements for low-connectivity regions.
- **Adaptive thresholds:** Lower initial caps for new users; graduates as trust builds; avoids early “walling off.”
- **Localization:** Mission content, instructions, and dispute flows are localized; host onboarding tailored by region.

8.10 Risk Matrix

Risk	Description	Likelihood	Impact	Mitigation
Gaming via repetition	Users spam low-value tasks	Medium	Medium	Repetition decay λ , per-period caps, template cooldowns
False positives	Legit missions flagged	Low–Med	Medium	Appeals with SLA, second-line adjudication, metric guardrails
Host integrity drift	Lax verification over time	Medium	High	Host trust scoring, random audits, slashing & re-certification
Privacy regressions	Over-collection creep	Low	High	Data minimization, off-chain storage, on-chain hashes only
Regional policy shocks	New KYC/NGO rules	Low–Med	High	Config toggles per region, tiered attestations, timelock updates
Score inequality	HII favors certain demographics	Medium	Medium	Balanced weights, alternative templates, fairness audits

8.11 Creative Framework — “The Impact Altimeter”

Visualize HII as an **altimeter** in a community aircraft. Engines (E/V/A missions) add altitude; thin air (repetition) reduces thrust; storms (fraud attempts) trigger instrument checks; air-traffic

control (governance) sets flight levels; the black box (indexer + events) records everything. The goal isn't just to climb fast—it's to **fly safely and land precisely** where impact is provable.

8.12 Quality Gate

- **Fact coherence:** HII inputs, decay, caps, and audit sampling align with mission templates and rewards constraints.
- **Abuse-resistance:** Each major exploit path has **two layers** of defense (signal + policy).
- **Contrarian review (brief):**
 - *Objection:* Strong anti-fraud can deter honest users. **Answer:** Default friction is low; heavy checks are adaptive and mostly invisible to good actors.
 - *Objection:* Single score hides nuance. **Answer:** HII is decomposable; dashboards expose per-type contributions and appeal outcomes.
 - *Objection:* Host trust bias. **Answer:** Trust is recalculated from outcomes; slashing and re-certifications keep hosts honest.

Chapter 9 — Peace Store & Commerce

9.1 Executive Summary

The Peace Store converts **verified impact** into **everyday utility**. It is a marketplace where HII (Human Impact Index) unlocks **tiered discounts, access, and perks**, while rigorous **anti-wash controls** keep GMV (gross merchandise value) honest. Commerce here is not an afterthought; it's the demand surface that closes the flywheel: missions → verified proofs → rewards → **real spending** → treasury revenues → (profit-conditioned) buybacks. This chapter details the Store's commercial model, merchant lifecycle, discount elasticity, risk controls, and reporting so that partners and auditors can trust the numbers—and users feel the value.

9.2 Key Findings

1. **Utility, not hype:** The Store converts tokens and reputation into tangible benefits—driving durable, non-speculative demand.
2. **GMV you can believe:** A **de-wash** pipeline and dispute flows ensure reported GMV aligns with real buyer–seller activity, not looped transactions.
3. **Merchant-first economics:** Repeat-merchant rate, reliable settlement, and transparent fee splits matter more than vanity volumes.
4. **Parameterization over guesswork:** Discount elasticity, velocity caps, and category ceilings are all DAO-tunable in the Config Registry, with machine-readable diffs.
5. **Safety by layers:** Risk scoring, graph anomaly detection, sampling audits, and cool-downs create a commerce perimeter that is hard to game and easy to monitor.

9.3 Methodology

- **Value-chain breakdown:** Modeled flows between **users (HII-tiered), merchants, hosts/NGOs, and treasury**, then assigned measurable KPIs and guardrails to each handoff.

- **Abuse catalogue** → **control mapping**: For each abuse pattern (wash loops, coupon arbitrage, collusion), defined at least two orthogonal controls (policy + signal).
- **Elasticity calibration**: Introduced discount bands by HII tier and product category; tested scenarios where GMV grows but net contribution per order degrades, and designed feedback levers.
- **Accounting clarity**: Defined canonical *Clean GMV* and *Net Contribution* so financial and token reports reconcile without manual heroics.

9.4 Commercial Model

9.4.1 Roles and Incentives

- **Users**: Spend rewards and/or \$PEACE to obtain goods/services with **HII-tiered** benefits.
- **Merchants**: Acquire incremental demand & brand halo; accept transparent fees and optional settlement in fiat/crypto.
- **Hosts/NGOs**: Earn a share on mission-linked commerce (when applicable) to align grassroots actors with healthy commerce.
- **Treasury**: Collects Store fees to fund operations; **only realized net profit** is eligible for buybacks (outside the Store's scope but relevant for budgeting).

9.4.2 Fee & Discount Surfaces

- **Base fee** (e.g., 2–5%) varies by category and risk tier.
- **HII-tiered discount** (e.g., 3–15%) increases with verified impact; caps apply per user, per period, and per category.
- **Promotional overlays** are time-boxed and rate-limited (e.g., “Back-to-School” bundles).
- **Settlement options**: Merchants can choose stable settlement (fiat/stablecoin) or partial \$PEACE payouts with programmatic vesting.

9.4.3 Clean Revenue Logic

Net Contribution = Clean GMV × (Base Fee – Discount Subsidy) – (Risk & Ops Reserves).

- *Clean GMV* excludes flagged orders and probabilistically discounts high-risk segments until cleared.
- *Discount Subsidy* is accounted per HII tier with budget ceilings (cannot exceed allocated subsidy pool).
- *Risk & Ops* reserve funds disputes, chargebacks, and fraud loss provisions.

9.5 Merchant Lifecycle

1. Onboarding & Tiering

- Submit KYC/registration + category details; auto-score initial risk.
- Probation Tier: lower discount ceilings, enhanced monitoring, and delayed settlements (e.g., T).
- Graduate to **Standard** and **Preferred** tiers based on on-time fulfillment, dispute rate, and repeat-merchant KPI.

2. Catalog & Compliance

- Category taxonomy with prohibited items (e.g., age-restricted goods) and regional policies.
- Listings include price, stock, refund terms, delivery windows; all changes logged.

3. Orders & Settlements

- `place_order(user, merchant, amount, hhi_tier)` applies discount policy and writes an immutable order record.
- **Hold & Flag**: suspicious patterns route to review; settlements pause until cleared.
- **Fee Splits**: treasury / host / ecosystem funds, all transparent and machine-readable.

4. Feedback & Disputes

- Two-sided ratings; disputes have a bounded SLA (e.g., ≤72h) with evidence upload; outcomes logged and used to recalibrate risk scores.

- **Chargeback playbook** with merchant coshare reduces moral hazard.

9.6 Anti-Wash & Risk Controls

9.6.1 Detection Primitives

- **Velocity caps:** per user and per merchant (hour/day/week); tighter in Probation tier.
- **Loop detection:** graph analysis for user ↔ merchant ↔ host cycles, shared device fingerprints, and repeated basket signatures.
- **Anomaly features:** sudden spikes vs. historical baseline, too-perfect redemption timing, orders just under cap thresholds.
- **Category heuristics:** tighter caps for high-risk categories (e.g., gift cards, easily resellable goods).

9.6.2 Policy Levers

- **Discount elasticity:** If Clean GMV rises but **Net Contribution** falls below a floor, reduce tier discounts automatically next epoch.
- **Per-category ceilings:** Limit redemptions in known-risk categories; cooldowns after max utilization.
- **Settlement throttles:** Increase settlement latency when risk indicators escalate.
- **Merchant probation & slashing:** Downgrade tiers, increase fees, or withhold fee shares for repeat offenders.

9.6.3 Dispute & Appeals

- Standard outcomes: **Uphold**, **Partial**, **Reverse**.
- Merchant win/lose ratios feed tiering; serial consumer abusers receive **consumer probation** (reduced discounts, extra checks).
- Evidence hashes are anchored on-chain; PII remains off-chain.

9.7 KPIs & Definitions (Publish by Default)

- **Clean GMV (Store)** = $\text{GMV} - \text{Flagged/Invalidated Orders (after de-wash)}$.
- **De-wash Rate** = $\text{Flagged Value} / \text{GMV}$ (target < x%).
- **Net Contribution** = $\text{Clean GMV} \times (\text{Fee} - \text{Subsidy}) - \text{Risk \& Ops Reserve}$.
- **Repeat-Merchant Rate** = share of merchants with monthly transactions in ≥ 3 consecutive months.
- **Discount Elasticity** = $\Delta \text{Clean GMV} / \Delta \text{Avg Discount (per tier/category)}$.
- **Dispute Ratio** = $\text{Disputed Orders} / \text{Clean Orders}$; **Resolution SLA** share met ($\leq 72\text{h}$).
- **User Spend Depth** = median orders per active user per month (by HII tier).
- **Coverage Assist** = treasury contribution from Store vs. reward expense (viewed in Tokenomics dashboards).

Publishing cadence: near-real-time tiles + monthly signed digest (CSV + on-chain hash). KPIs must reconcile with ledger events (orders, flags, settlements).

9.8 Economic Scenarios

1. “Promotional Overheat”

- *Symptom:* GMV rises sharply; Net Contribution turns negative.
- *Response:* Down-shift discount elasticity next epoch; narrow high-elasticity categories; raise velocity caps for **clean** segments only; publish a mid-cycle note explaining changes.

2. “Probation Drag”

- *Symptom:* Many new merchants stuck in probation, depressing selection.
- *Response:* Introduce curated bundles with vetted partners; time-boxed “probation booster” with slightly higher caps under strict monitoring; share best practices to accelerate graduation.

3. “Loop Farming”

- *Symptom*: Closed loops between a handful of users and a merchant spike at end of epoch.
- *Response*: Auto-flag, throttle settlement, re-price discounts for involved HII tiers; audit host/NGO ties; if collusion proven, slash merchant fee share and mark participants for probation.

4. “Seasonal Shift”

- *Symptom*: Genuine spike (e.g., back-to-school) but risk features also jump.
- *Response*: Temporarily raise category ceilings for vetted merchants while increasing sampling rates; maintain transparency through weekly KPI footnotes.

9.9 User Experience (Funnel, Not Friction)

- **Browse → Select → Apply HII Tier → Pay** in ≤ 3 taps inside the MiniApp.
- **Transparent pricing**: show base price, HII discount, and final net; explain “why this tier.”
- **Receipts**: on-chain order hash + human-readable invoice; link to dispute center.
- **Nudges**: personalized “next-tier” hints (e.g., “Complete 2 verified missions to unlock +3%”).
- **Privacy**: payment PII stays off-chain; only hashes + classifications are committed on-chain.

9.10 Compliance & Category Governance

- **KYC tiers** for merchants and, where required, for buyers.
- **Regional toggles** (e.g., sanctions, age-gating) in the Config Registry; all changes timelocked and diffed.
- **Prohibited goods list** maintained with versioning; exceptions require “Critical” proposals.
- **Refund/return policy** standardized by tier; deviations must be disclosed in listing metadata.

9.11 Risk Matrix

Risk	Description	Likelihood	Impact	Mitigations
Wash trading	Synthetic loops inflate GMV	Med	High	Velocity caps, loop detection, de-wash pipeline, settlement throttles
Coupon arbitrage	Resale of discounted goods	Med	Med High	Category ceilings, cooldowns, risk tiering, partner SKUs (non-resellables)
Chargebacks/disputes	Post-settlement fraud or poor service	Med	Med	SLAs, reserves, merchant tiering, evidence anchoring
Subsidy overspend	Discounts exceed budget value	Med	High	Discount elasticity, subsidy ceilings, auto down-shift rules
Compliance breach	Wrong-way sales (jurisdiction/rules)	Low–Med	High	Category governance, regional toggles, audits, logs
Merchant churn	Poor economics reduce supply	Med	Med	Predictable settlement, fee transparency, merchant success ops

9.12 Creative Framework — “The Harbor Model”

See the Store as a **harbor**:

- **Lighthouses** (risk scores) guide ships in; **pilots** (probation rules) escort newcomers; **harbor fees** (base fees) keep services running.
- **Tides** (discount elasticity) rise and fall with seasonality; **breakwaters** (anti-wash controls) blunt artificial waves.

- **Manifests** (order logs) and **customs** (compliance toggles) keep trade honest. A good harbor isn't the busiest—it's the **safest path to real trade**, where captains return because the math works.

9.13 Quality Gate

- **Coherence:** Every price move and risk decision is parameterized and logged; no manual exceptions invisible to the ledger.
- **Economics:** Clean GMV and Net Contribution make subsidy spending accountable; elasticity rules correct drift automatically.
- **Contrarian review (brief):**
 - *Objection:* Tight anti-wash rules may deter honest users. **Answer:** Controls scale with risk tier; Standard/Preferred merchants see **lower friction** than Probation.
 - *Objection:* Discounts can train users to wait for promos. **Answer:** HII-tiered benefits are **earned**, not seasonal; promos are time-boxed and rate-limited.
 - *Objection:* Merchant churn could spike if disputes rise. **Answer:** Predictable SLAs, evidence anchoring, and tiered reserves reduce surprises; merchant success ops focus on repeat-merchant health.

Chapter 10 — Risk, Legal, and Privacy

10.1 Executive Summary

Peace Soldiers is engineered to be **safe-by-design**: risks are identified early, mapped to explicit controls, and instrumented so that evidence—not promises—proves compliance over time. The legal perimeter focuses on three cores: (1) **Financial/market conduct** (token utility, disclosures, treasury discipline), (2) **Data protection & privacy** (data minimization, on-chain hash/off-chain PII split, DPIA, incident playbooks), and (3) **Platform integrity** (KYC/AML tiers where required, sanctions screening, content/merchant policy, dispute/chargeback flows). Governance turns policy into code through a **Config Registry**, **timelocks**, and **SOPs** so critical toggles are transparent and revertible. The result is a system built to operate under scrutiny—technical, legal, and social.

10.2 Key Findings

1. **Risk is a lifecycle, not a checklist.** Threats evolve; therefore, controls must be parameterized, versioned, and observable.
2. **Utility-first token model reduces securities exposure.** No revenue-share promises; buybacks are **profit-conditioned** and bounded; value accrues from use, access, reputation, and governance utility.
3. **Privacy is an architecture choice.** On-chain stores **hashes and verdicts**; sensitive artifacts remain off-chain with deletion pathways, regionalization, and least-privilege access.
4. **Compliance at the edge.** KYC/AML and sanctions checks are **jurisdiction-toggleable**; merchant categories and user tiers can be scoped per region without redeploying contracts.
5. **Prove it.** Logs, audits, and monthly signed digests (anchored on-chain) transform obligations into verifiable evidence.

10.3 Methodology

- **Top Risks Inventory:** Legal, operational, market, counterparty, privacy, and reputational risks enumerated against Chapter 4–9 components.
- **Control Mapping:** Each risk receives at least **two** orthogonal mitigations (policy + technical).
- **Evidence Design:** For every control, define the **artifact** (event, log, signed report) that proves its use.
- **Review Cadence:** Quarterly risk review; incident retros within 72h; annual external audits (security + privacy + financial processes).

10.4 Deep-Dive: Risk Domains & Controls

10.4.1 Market & Treasury Risk

Threats: reflexive emissions, treasury depletion, price manipulation.

Controls:

- **Coverage-first** rule: buybacks off unless *realized net profit* and coverage $\geq$ threshold.
- TWAP windows and oracle freshness checks; per-window notional caps. (See Annex E: Oracle Providers & Max-Age table.)
- **Budget ceilings** on rewards; claim-time burn; mission multipliers bound by registry.
Evidence: *BuybackQueued/Executed* events, coverage calculations in monthly digest, config diffs.

10.4.2 Governance & Conduct Risk

Threats: vote capture, proposal stuffing, hurried changes.

Controls:

- Identity caps, optional HII-weighting, quadratic dampening.
- **Classed proposals** with quorums and timelocks; single-subject rule; disclosure packs.
- Public diffing of **Config Registry**; rollback plans for Critical changes.
Evidence: proposal logs, diffs, turnout metrics, timelock windows.

10.4.3 Operational & Fraud Risk

Threats: mission farming, store wash-trading, merchant abuse.

Controls:

- Multi-signal verification, randomized audits, penalties/appeals with SLA.
- Store **de-wash** pipeline, velocity caps, loop detection, tiered settlements.
Evidence: verification rates, dispute outcomes, flagged-order ledger, merchant tier changes.

10.4.4 Counterparty & Merchant Risk

Threats: poor service, chargebacks, category non-compliance.

Controls:

- Tiered onboarding (Probation → Standard → Preferred), fee/discount guardrails by category.
- Dispute SLAs and reserve funds; parameterized **regional toggles**.
Evidence: settlement logs, reserves accounting, category lists with versioning.

10.4.5 Privacy & Data Protection

Threats: over-collection, data breaches, unlawful processing.

Controls:

- **Data minimization:** Only hashes/verdicts on-chain; PII off-chain with TTL and deletion paths.
- **DPIA & ROPA:** Data Protection Impact Assessment; Records of Processing Activities maintained.
- **Access control:** Role-based access; encryption at rest/in transit; audit trails for all PII access.
- **Localization:** Regional data residency where required; geofenced processing.
- **Incident Response:** Detection → containment → notification within policy windows; post-mortem on-chain hash.
Evidence: DPIA versions, access logs, deletion tickets, incident reports.

10.4.6 Compliance & Financial Crime

Threats: AML breaches, sanctioned-party exposure.

Controls:

- **KYC/AML tiers** where the law requires; sanctions screening; enhanced due diligence for high-risk categories.
- **Transaction monitoring:** rules + anomaly detection; freeze/flag flows pending review.
- **Jurisdictional scoping:** Features toggled via Config Registry to align to local rules without touching core logic.
Evidence: KYC/EDD audit logs (off-chain), sanctions check attestations, freeze/unfreeze events.

10.4.7 Consumer & Advertising Risk

Threats: misleading claims, unfair practices, financial-promotion violations.

Controls:

- Clear **utility** statements; no earnings promises; standardized disclosures.
- Region-specific disclaimers; age/suitability gates where applicable.
Evidence: versioned Terms, policy snapshots, release notes.

10.4.8 IP & Content Policy

Threats: unauthorized content, trademark/copyright claims.

Controls:

- Merchant content warranties; takedown process; repeat-infringer policy.
- Asset fingerprinting for common infringements.
Evidence: takedown tickets, repeat-offender listings, policy versions.

10.5 Compliance Matrix (What, How, Evidence)

Area	Control	Mechanism	Evidence Artifact
Rewards solvency	Epoch ceilings	On-chain registry param	<code>ConfigUpdated</code> , budget tables
Net-profit buybacks	Coverage & profit TWAP + oracle freshness gates		<code>BuybackExecuted</code> , monthly digest
Identity integrity	One passport per Soulbound cNFT + human	device heuristics	<code>PassportIssued/Revoked</code> , duplicate reports
Mission verification	Multi-signal + audits	GPS/media/host + sampling	<code>MissionFinalized</code> , audit outcomes
Store integrity	De-wash pipeline	Loop detection + velocity caps	<code>OrderFlagged/Settled</code> , Clean GMV
Data protection	Minimize + deletion path	Off-chain PII + on-chain hash	DPIA, deletion logs, access trails
AML/sanctions	Tiered KYC + screening	Regional toggles	KYC audit logs, freeze events
Governance change	Timelock + diffs	Classed proposals	<code>ProposalQueued/Executed</code> , diffs
Incidents	SOP drills & notes	EmergencyStop toggles	Incident notes, post-mortem hash

10.6 Privacy Architecture (PII-Sparse by Design)

- **Data split:**
 - **On-chain:** hashes, timestamps, verdict codes, parameter changes.
 - **Off-chain:** media, GPS artifacts, IDs, merchant documentation; encrypted, access-logged, and region-scoped.
- **Retention:** Mission artifacts retained only as long as needed for verification/dispute windows; merchants' KYC per legal minimums.
- **User rights:** Access/export/erasure requests honored where lawful; responses logged.
- **Testing:** Privacy unit tests (no PII accidentally logged); synthetic datasets for ML/QA.

10.7 Incident Response & Business Continuity

Triggers: exploit indicators, oracle failure, data breach suspicion, sanctions updates, governance anomalies.

Playbook:

1. **Detect & Triage** (classify severity)
2. **Contain** (toggle EmergencyStop per surface; rotate keys if needed)
3. **Notify** (internal, community, regulators where required)
4. **Remediate** (patch, config changes, compensation if applicable)
5. **Document** (forensic package; lessons learned)
6. **Publish** (summary with on-chain hash; follow-up checks)

BCP (Continuity): geo-redundant infra; cold-start runbooks; council quorum backups; indexer re-ingestion scripts.

10.8 Disclosures & Document Set (Versioned)

- **Whitepaper** (this document) — utility description; not an offer of securities.

- **Terms of Use** — eligibility, acceptable use, disclaimers, dispute forum/venue.
- **Privacy Notice** — purposes, retention, rights, contacts.
- **Merchant Agreement** — fees, settlement, reserves, category rules, SLA.
- **Open Methodology** — KPI definitions, Clean GMV formula, buyback SOP excerpt.
- **Risk Factors** — operational, market, legal, technology, and governance.

All documents carry semantic versions; changes appear in release notes and are hashed on-chain.

10.9 Risk Matrix (Consolidated)

Risk	Description	Likelihood	Impact	Primary Mitigations
Treasury depletion	Buybacks without profit/runway	Low	High	Profit-only gate, coverage threshold, TWAP/oracle freshness
Governance capture	Concentrated voting power	Med	High	Identity caps, HII-weight, quorums, timelocks
Mission fraud	GPS/media spoofing, host collusion	Med	High	Multi-signal, audits, penalties, host trust scoring
Store wash	Self-dealing loops inflate GMV	Med	Med–High	De-wash pipeline, velocity caps, tiered settlement
Data breach	PII exposure	Low	High	Encryption, least-privilege, monitoring, IR playbook
Regulatory shock	New KYC/AML or sanctions	Low–Med	High	Jurisdiction toggles, scoped rollouts, legal review cadence
Misleading comms	Utility misstatements	Low–Med	Med–High	Standardized disclosures, approvals, audit trail

10.10 Creative Framework — “Compliance as a Living Ledger”

Imagine compliance as a **living ledger**: every obligation is a line item with a debit (risk) and a credit (control). Entries post in real time—**events, diffs, hashes, reports**—so the ledger always balances or flags variance. Auditors don’t take our word; they **replay the ledger**. When regulation shifts, we don’t rewrite history—we **append**: new obligations, new credits, new evidence. That’s how the system stays current without losing its memory.

10.11 Quality Gate

- **Coherence check:** Each high-materiality risk maps to ≥ 2 controls and a concrete evidence artifact.
- **Legal posture:** Utility-first token language; no earnings promises; profit-conditioned buybacks; jurisdictional scoping via configurable toggles.
- **Contrarian review (brief):**
 - *Objection:* “Config toggles can be abused off-chain.” **Answer:** All toggles are on-chain, timelocked, and diffed; council cannot bypass DAO routes.
 - *Objection:* “Data minimization limits analytics.” **Answer:** Aggregations rely on hashes and metadata; sensitive artifacts remain fetchable under access controls.
 - *Objection:* “Profit-only buybacks reduce market support.” **Answer:** That is the point—support must be **earned by operations**, not optics.

Chapter 11 — Roadmap & Delivery

11.1 Executive Summary

This roadmap converts the architecture into an execution plan with **crisp milestones, accountable owners, budget envelopes, and verification KPIs**. Delivery is staged across four waves—**Bootstrap, Closed Beta, Open Beta, Global Launch**—with **progressive decentralization** and **evidence gates** at every transition. We treat dates as **contracts with telemetry**: each milestone is paired with success metrics, rollback criteria, and an audit artifact so stakeholders can verify—not assume—readiness.

11.2 Key Findings

1. **Evidence-gated progress** beats deadline theater. Each phase ships with checkable artifacts: passing test suites, security/privacy reports, public KPI tiles, and on-chain snapshot hashes.
2. **Thin slices, not big bangs**. We deliver in minimal, value-complete increments—identity → missions → rewards → store → governance—so feedback and fraud signals shape the next slice.
3. **Two clocks, one plan**. Product clock (shipping value) and Risk clock (closing attack surfaces) run in parallel; transitions require both to be green.
4. **Progressive decentralization** is earned by reliability: upgrade keys and treasury powers move to the DAO only after quantifiable stability windows.
5. **Telemetry is a feature**. KPI pipelines, playbooks, and public digests are first-class scope, not “post-launch” items.

11.3 Methodology

- **Backward planning from Global Launch**: define the final state (all modules live, SOP drills passed, DAO online), then derive interim “Definition of Done” per phase.

- **Workstream slicing:** Access, Identity, Proof, Incentives, Commerce, Governance, and Data each have phase-appropriate deliverables, owners, and test evidence.
- **Risk-led gating:** Every phase has “Ship / Hold / Rollback” criteria tied to quantitative thresholds (e.g., verification rate, incident-free days, coverage months).

11.4 Phased Roadmap (Milestones, Owners, Evidence)

Phase 0 — Bootstrap (Weeks 0–4)

Scope:

- Config Registry v1; Passport mint (dev-only); Missions templates v0 (E/V/A skeleton); Rewards accrual/claim happy path; Indexer MVP; KPI tiles (MVR, claims).
Owners: Platform Eng (lead), Data Eng, Security.
Evidence gates:
 - 90%+ unit/prop tests pass;
 - First public dashboard (read-only) with on-chain hash of snapshot;
 - SOP drills (EmergencyStop toggles) executed in staging with logs.

Phase 1 — Closed Beta (Weeks 5–10)

Scope:

- MiniApp onboarding; limited Passport issuance; Missions verification pipelines (signals & sampling); Rewards claim with **burn bps**; Store probation merchants; DAO “Operational” class only.
Owners: Product & Mobile, Platform Eng, Risk/Compliance.
Evidence gates:
 - Mission Verification Rate (MVR) $\geq 85\%$; disputes SLA $\leq 72h$ met $\geq 95\%$;
 - Reward coverage ≥ 12 months at current run-rate;
 - No P1 security incidents in 21 consecutive days;
 - Monthly KPI digest (CSV + on-chain hash).

Phase 2 — Open Beta (Weeks 11–20)

Scope:

- Expand merchants; enable Economic-class proposals; Anti-wash pipeline live; Buyback module in **monitor** mode (no execution); DAO delegation UI; localization in 2–3 priority locales.

Owners: Commerce, Governance, Data, Legal.

Evidence gates:

- De-washed GMV growth $\geq 15\%$ MoM; dispute ratio $\leq 2\%$ with $\geq 95\%$ SLA;
- Governance turnout $\geq$ threshold; config diffs visible;
- Incident log: zero unresolved P1s; post-mortems for all P2s;
- Privacy DPIA v1 signed; access logs sampled.

Phase 3 — Global Launch (Weeks 21–32)

Scope:

- Profit-conditioned buybacks ON under TWAP + coverage guards; DAO “Critical” class enabled; preferred merchant tiers; grants pilot; public audit reports (security + privacy) summarized.

Owners: Treasury, Governance Council, Security, Commerce.

Evidence gates:

- Coverage months $\geq$ threshold; oracle freshness alarms green 30-day rolling; (See Annex E: Oracle Providers & Max-Age table.)
- Buyback SOP receipts public;
- Governance concentration (Herfindahl) $\leq$ target;
- Uptime $\geq 99.5\%$; dashboard freshness $\leq 60s$ lag.

Phase 4 — Progressive Decentralization (Weeks 33+)

Scope:

- Hand-off upgrade keys; elevated quorums; stewardship elections; external data mirrors; open data schema docs.

Owners: Governance, Community, Data.

Evidence gates:

- 90-day stability window (no emergency halts);
- External replicators validate KPIs from logs;
- Key rotations executed and announced on-chain.

11.5 Workstreams, Deliverables, and Acceptance

Workstream	Phase-Scoped Deliverables	Acceptance / Evidence
Access (MiniApp)	Onboarding, wallet connect, mission wizard, receipts	$p95 \leq 1.2s$, crash rate $\leq 0.5\%$, signed payload verification logs
Identity (Passport)	SBT mint/revoke, flags, duplicate heuristics	PassportIssued/Revoked events; duplicate rate trend ↓
Proof (Missions)	Templates, signals, audits, appeals	$MVR \geq \text{target}$; audit sampling coverage; appeal SLA met
Incentives (Rewards/Treasury)	Accrual → claim (burn bps), budget ceilings, buyback monitor → execute	Coverage months $\geq \text{target}$; buyback receipts; RewardClaimed receipts
Commerce (Store)	Merchant registry, anti-wash pipeline, fee splits	Clean GMV ↑; de-wash rate $\leq x\%$; settlement SLA met
Governance (DAO)	Proposal classes, caps, HII-weight mode, timelocks	Diffs public; turnout $\geq \text{threshold}$; concentration $\leq \text{target}$
Data/Telemetry	Indexer, dashboards, public digests	0 data gaps > 5 min; monthly CSVs hashed on-chain
Security/Privacy	Threat model, DPIA/ROPA, drills	Zero P1s; drill evidence; privacy tests pass

11.6 Dependencies & Critical Paths

- **Config Registry** precedes every module's tuning; early lock prevents "hidden flags."
- **Indexer & KPIs** must be ready before Closed Beta to make evidence-gated decisions.
- **Store anti-wash** must precede Global Launch to prevent GMV distortion.
- **Legal artifacts** (Terms, Privacy, Merchant Agreement) must be versioned before Open Beta.
- **Oracle feeds** and **buyback venues** must be integration-tested before enabling execution.

11.7 Team Structure, RACI, and Budget Envelopes

Core Pods (RACI):

- **Platform Eng (A/R)** — contracts, registry, treasury; **Security (C)**; **Governance (I)**
- **Product & Mobile (A/R)** — MiniApp UX; **Platform (C)**; **Data (I)**
- **Data Eng (A/R)** — indexer, KPIs; **All (C)**
- **Commerce (A/R)** — merchant ops, disputes; **Legal/Risk (C)**
- **Governance & Legal (A/R)** — DAO process, documents, disclosures; **Security (C)**
- **Security & Privacy (A/R)** — threat model, DPIA, drills; **All (I)**

Budget envelopes (illustrative % of total for first 9 months):

- Engineering 38–45% | Security/Privacy 8–12% | Data/Infra 12–16% | Commerce Ops 10–14% | Governance/Legal 8–12% | Community/Support 6–10% | Contingency 5–8%.

11.8 OKRs (Quarterly)

Objective 1 — Ship a reliable Closed Beta

- KR1: MVR $\geq 85\%$ with $\leq 2\%$ false-positive rate.
- KR2: Reward coverage ≥ 12 months; burn_bps tuned $\leq \pm 25$ bps variance target.
- KR3: Dashboard freshness ≤ 60 s; 100% monthly digest published.

Objective 2 — Prove commerce utility

- KR1: Clean GMV $\geq \$X$ in Q; de-wash rate $\leq y\%$.
- KR2: Repeat-merchant rate $\geq 60\%$; dispute SLA met $\geq 95\%$.
- KR3: Discount elasticity within $\pm Z$ band of Net Contribution target.

Objective 3 — Governance readiness

- KR1: ≥ 2 successful Economic proposals with diffs & rollback plans.
- KR2: Turnout $\geq$ threshold; Herfindahl $\leq$ target.
- KR3: SOP drills (EmergencyStop, buyback halt) executed and published.

11.9 QA & Launch Readiness (LTR)

LTR Checklist (extract):

- Unit/prop tests $\geq 90\%$; fuzz suite passes; gas/compute budgets within target.
- Security audit issues (High/Critical) remediated; residual risks documented.
- DPIA signed; access logs sampled; deletion/erasure tested.
- Incident runbooks rehearsed; paging on-call rotation active.
- Buyback guardrails validated (coverage/profit/oracle freshness). (See Annex E: Oracle Providers & Max-Age table.)

- Public docs (Terms/Privacy/Merchant) live; versions hashed on-chain.
- “Known Limitations” page published.

Rollback policy: any P1 in the last 7 days → hold launch; 2× P2 in 72h → launch committee review; bad oracles → disable buybacks.

11.10 Communications Plan

- **Closed Beta:** invite-only cohorts; weekly changelog; “How to read KPIs” explainer.
- **Open Beta:** monthly AMAs; merchant spotlights; transparency notes on de-wash and dispute metrics.
- **Global Launch:** independent audit summaries; community vote walkthrough; “What changes after decentralization” guide.
- **Crisis comms:** templated incident notices with impact scope, user actions (if any), and ETA; post-mortems published with hashes.

11.11 Risks to the Plan & Mitigations

Risk	Symptom	Mitigation
Verification backlog	Appeals exceed SLA	Increase sampling automation; surge reviewers; prioritize templates causing false positives
Merchant shortfall	Low catalog breadth	Add onboarding bounties; curated bundles; partner BD sprint
KPI integrity drift	Dashboard vs. ledger mismatch	Data audit jobs; external mirrors; public query fingerprints
Governance apathy	Low turnout	Default delegation UX; reputation badges; reminder cadence
Scope creep	Slipping milestones	Freeze non-critical features; guard “evidence gates” as hard stops

11.12 Creative Framework — “Dual-Lane Highway”

Execution runs on **two lanes**:

- **Lane A (Value Lane):** ships user-visible utility each sprint.
- **Lane B (Proof Lane):** ships the evidence—tests, KPIs, audits, SOP drills—that lets us keep shipping.
Green lights require **both** lanes to arrive at the same intersection. If Value arrives without Proof, we **stop**. If Proof arrives without Value, we **re-sequence**. This keeps speed without sacrificing safety.

11.13 Quality Gate

- **Fact coherence:** Every phase has measurable gates tied to the control surfaces defined earlier (registry, SOPs, KPIs).
- **Operational realism:** Capacity, SLAs, and security/privacy tasks are scheduled—not hand-waved.
- **Contrarian review (brief):**
 - Objection: “Evidence gates slow growth.” **Answer:** They prevent expensive rework and reputational losses; throughput increases once baselines stabilize.
 - Objection: “Merchant ramp will lag.” **Answer:** Probation tier + curated bundles maintain quality while widening supply.
 - Objection: “DAO readiness is subjective.” **Answer:** We bind it to objective signals (stability window, turnout, concentration index, audit close-out).

Chapter 12 — Team & Organization

12.1 Executive Summary

World-class systems require world-class teams—disciplined, cross-functional, and measured by evidence, not vibes. This chapter codifies how Peace Soldiers organizes people, decisions, and accountability to deliver a **secure, audit-ready, product-obsessed** platform. We define the operating model (pods and councils), role charters, hiring plan, performance system, decision rights, and vendor governance—so execution scales without losing judgment.

12.2 Key Findings

1. **Pod + Council model:** Execution flows through **product pods** (Access, Identity, Proof, Incentives, Commerce, Governance, Data) while **risk councils** (Security/Privacy, Treasury, Compliance) set guardrails.
2. **Decision clarity > speed illusions:** We use **DRIs** (Directly Responsible Individuals), **RAPID/RACI** hybrids, and **timelocks** for high-impact decisions to avoid “fast mistakes that last.”
3. **Evidence-first leadership:** Every pod commits to telemetry: OKRs, SLAs, and a monthly **Signed Digest**. Promotions and funding decisions hinge on these artifacts.
4. **Two-track capacity:** “Value Lane” ships user features; “Proof Lane” ships audits, SOP drills, and KPI integrity—both must pass gates to progress (continuity from Chapter 11).
5. **Culture of measured boldness:** We reward **impact density** (paired product+proof output), ethical judgment, and user empathy; we sunset low-yield activities aggressively.

12.3 Methodology

- **Back-solved** from the product’s control surfaces (Ch. 4–9) to minimal roles needed to own them end-to-end.
- **Benchmarked** talent stacks from high-scale consumer crypto, fintech risk, and marketplace ops.

- **Stress-tested** org design against failure modes: security incidents, governance capture, fraud spikes, and runaway scope.

12.4 Org Topology (Pods, Councils, and Lines)

12.4.1 Product Pods (P&L-adjacent)

Each pod is cross-functional with an embedded PM, Tech Lead, and Data partner. Pods own a **control surface**, a **backlog**, and **phase gates**.

- **Access (MiniApp):** Onboarding, wallet connect, mission wizard, receipts.
- **Identity (Passport):** SBT issuance/revocation, duplicate heuristics, KYC tiers.
- **Proof (Missions):** Template registry, verification pipeline, appeals.
- **Incentives (Rewards/Treasury):** Accrual → claim (burn bps), budget ceilings, buyback module (with Treasury Council).
- **Commerce (Store):** Merchant lifecycle, anti-wash, fee/discount policy.
- **Governance (DAO):** Proposal classes, timelocks, voting UX, Config Registry flows.
- **Data & Observability:** Indexer, dashboards, public digests, external mirrors.

Pod cadence: Weekly sprint; bi-weekly demo; monthly **Evidence Pack** (KPI tiles + on-chain hash).

12.4.2 Risk & Policy Councils (Guardrail owners)

- **Security Council:** Threat model, secure SDLC, incident response; signs off Critical releases.
- **Privacy Council (DPO-led):** DPIA/ROPA, data minimization, deletion tests, regulator interface.
- **Treasury Council:** Profit definition, coverage thresholds, buyback venue policy; reconciles monthly.
- **Compliance Council:** KYC/AML, sanctions, merchant policy, regional switches; runs table-top drills.

- **Ethics & Integrity Board:** Reviews sensitive missions/partners; maintains prohibited list; mediates gray areas.

Council cadence: Monthly policy review; quarterly drills; emergency huddles as needed.

12.5 Decision Rights & Escalation

Decision Area	Owner (A/R)	Consulted (C)	Informed (I)	Gate
Mission template change	Proof Pod	Security, Data	Governance Pod	Economic-class proposal if scoring weights shift
Burn bps / rewards ceiling	Incentives Pod	Treasury, Data	Governance Pod	Economic-class proposal + timelock
Buyback params execution	Treasury Council	Incentives, Data	Governance Pod	Critical-class proposal, TWAP guards
Merchant risk policy	Commerce Pod	Compliance, Security	Data	Council sign-off, staged rollout
Config Registry updates	Governance Pod	Security, All pods	All	Class-based quorums & timelocks
PII handling & retention	Privacy Council	Security, Data	All	DPIA sign-off; audit trail
Emergency halts	Security Council	Governance Chair	All	SOP engage → post-mortem hash

DRI contract: One named person per decision owns the outcome, artifacts, and communication.

12.6 Role Charters (Critical Roles)

- **Head of Platform Engineering:** Owns contracts, registry, and performance envelope; accountable for Critical releases; co-signs SOP drills.
- **Head of Product (MiniApp):** Conversion, retention, crash budget; UX for identity & missions; accessibility/localization.
- **Chief Security Officer (CSO):** Secure SDLC, vuln mgmt, incident response, pen-test coordination; approves Critical deploys.
- **Data Lead:** Indexer reliability, KPI correctness, query fingerprinting; maintains public schemas & digest pipeline.
- **Treasury Lead:** Coverage calc, P&L reconciliations, buyback SOP execution—the **profit gatekeeper**.
- **Privacy/DPO:** DPIA/ROPA, access logs, user rights workflows; audits “minimum data” discipline.
- **Head of Commerce:** Merchant growth & health, dispute SLAs, Net Contribution guardrails.
- **Governance Lead:** Proposal hygiene, turnout programs, delegate UX, registry diffing.
- **Risk Lead (Fraud/Compliance):** Anti-wash, sanctions watch, KYC tiers; runs red-team scenarios.
- **DevRel/Community Lead:** Docs, AMAs, moderator program, localization partners.

12.7 Hiring Plan (First 9–12 Months)

Wave 1 (Bootstrap, 0–3 months | ~12–16 heads)

- 2 Protocol/Smart Contract Eng, 2 Backend, 1 MiniApp Eng, 1 PM (Access), 1 PM (Proof), 1 Data Eng, 1 Security Eng, 1 Risk Analyst, 1 Designer, 1 Technical Writer, 1 DevOps/SRE.

Wave 2 (Closed → Open Beta, 4–8 months | +10–14)

- +1 Protocol, +1 Backend, +1 MiniApp, +1 Data (analytics), +1 QA Automation, +1 Compliance Ops, +2 Merchant Ops, +1 Treasury Analyst, +1 Localization PM, +1

Community/Support Lead.

Wave 3 (Global Launch, 9–12 months | +8–12)

- +1 Security Eng (AppSec), +1 Privacy Ops, +1 Governance Operations, +2 Merchant Success, +2 Partner Integrations, +1 Site Reliability, +1 Data Quality.

Talent bar: Prioritize **operators** who've shipped at scale; probe for **audit literacy, system thinking, and ethical judgment**. "Brilliant loners" without documentation discipline won't clear the bar.

12.8 Operating Cadence

- **Daily:** Pod standups (15 min); incident standup if Sev $\geq$ 2.
- **Weekly:** Sprint planning + demo; error budget review (SLOs, crash budgets, indexer lag).
- **Bi-weekly:** Release train; governance proposal review; config diff previews.
- **Monthly:** KPI & Signed Digest (CSV + on-chain hash), budget vs. coverage review, buyback eligibility check.
- **Quarterly:** Security tabletop, privacy drill (erasure & access), treasury reconciliation, roadmap re-baselining.
- **After any P1:** Post-mortem within 72h; action items with owners/dates; on-chain hash of summary.

12.9 Performance System (OKRs & Promotions)

OKR rubric (examples):

- Access: p95 latency < 1.2s; D30 retention $\geq$ 30%; signed payload verification $\geq$ 99.9%.
- Proof: MVR $\geq$ 90%, false-positive $\leq$ 2%, appeal SLA met $\geq$ 95%.
- Incentives: Coverage $\geq$ 12 months; zero budget overruns; burn_bps variance $\pm$ 25 bps of target.
- Commerce: De-washed GMV $\uparrow \geq$ 20% MoM; dispute ratio $\leq$ 2%; repeat-merchant $\geq$ 60%.

- **Governance:** Turnout $\geq$ threshold; Critical timelock $\geq$ window; registry churn within budget.
- **Data:** Dashboard freshness $\leq$ 60s; no data gaps $>$ 5 min; external mirror parity $\geq$ 99.5%.
- **Security/Privacy:** 0 unresolved High vulns $>$ 7 days; DPIA current; drills executed.

Promotion signals: multi-quarter **impact density**, quality of **evidence packs**, cross-org leadership, and reduction in **blast radius** per incident.

12.10 Security & Privacy Operations

- **Secure SDLC:** Threat modeling at design; mandatory code review; static/dynamic analysis; secrets scanning; supply-chain checks.
- **Vulnerability Mgmt:** Severity-based SLAs (High: 7d, Critical: 72h); public disclosure policy with timelines.
- **Detection & Response:** Centralized logging; anomaly alerts; 24/7 on-call rotation; tabletop drills.
- **Data Governance:** Access least-privilege; quarterly access recert; encryption at rest/transit; redaction in logs.
- **Privacy Ops:** DSAR process (access/erasure/export); retention schedules; audit of deletion outcomes; privacy tests in CI.
- **Third-Party Risk:** Vendor inventory, DPIA addenda, security questionnaires; kill-switches for non-compliant vendors.

12.11 Vendor & Partner Governance

- **Contracts:** Security exhibits, data processing addendums, uptime SLAs, escalation ladders.
- **Onboarding Checklist:** Pen-test summary, SOC2/ISO evidence (or compensating controls), breach notification windows, sandbox trial.
- **Monitoring:** Quarterly scorecards (SLOs, incidents, compliance attestations); **auto-renew blocked** unless score $\geq$ threshold.

- **Exit Plans:** Data return/erasure, key rotations, traffic cutover runbooks.

12.12 Community & Support

- **Moderation Playbook:** Clear code of conduct; tiered moderator privileges; evidence-based sanctions; appeal path.
- **Support SLAs:** First response ≤ 4 business hours; resolution targets by severity; knowledge base with release-synced articles.
- **Education:** “How to read KPIs,” “Why profit-conditioned buybacks,” “Appeals 101” content in multiple languages.
- **Recognition:** Non-monetary badges for constructive governance contributors and fraud reporters.

12.13 Diversity, Inclusion, and Ethics

- **Hiring:** Structured interviews; calibrated panels; diverse sourcing; bias-resistant rubrics.
- **Compensation:** Bands with published ranges; annual pay equity review.
- **Ethics:** Partner/missions review board; sunset criteria for ethically ambiguous programs; whistleblower channel with non-retaliation policy.
- **Accessibility:** WCAG-minded UI; low-bandwidth modes; screen-reader support; language localization strategy.

12.14 Risk Matrix (Org-centric)

Risk	Description	Likelihood	Impact	Mitigations
Key person risk	Over-reliance on few leads	Med	High	Two-in-a-box for critical roles; docs; succession plans
Scope creep	Pods chase shiny objects	Med	Med–High	Gate by Evidence Packs; freeze lists; roadmap council

Process theater	Evidence without outcomes	Low–Med	Med	Tie OKRs to user & coverage outcomes; audit KPI parity
Security drift	SDLC shortcuts near deadlines	Med	High	Release train gates; CSO veto; error budgets
Data integrity	Dashboards diverge from logs	Med	High	Query fingerprinting, external mirrors, monthly digest
Compliance lag	Regional rules outpace switches	Med	High	Compliance Council watchlist; config toggles; drills
Burnout / attrition	High pace, thin staff	Med	High	Sustainable on-call; rotation; focus sprints; wellness budget

12.15 Creative Framework — “The Orchestra Model”

Think of the organization as an **orchestra**:

- **Strings (Pods)** carry the melody—shipping features users hear.
- **Brass (Councils)** provide power and restraint—security, privacy, treasury.
- **Percussion (Data)** keeps time—KPIs and digests.
- **Conductor (Governance)** sets tempo and dynamics—proposal classes and timelocks. Great performances don’t come from louder instruments; they come from **score discipline**—when to enter, how loud to play, and when to rest. The score here is **evidence**.

12.16 Quality Gate

- **Coherence:** Every critical product surface has a named owner, council, cadence, and artifact.
- **Operability:** Decision rights and escalation routes are unambiguous; releases are gated by measurable proof.

- **Contrarian review (brief):**

- *Objection:* “Councils will slow shipping.” **Answer:** Councils set guardrails, not backlogs; gates are risk-based and time-boxed.
- *Objection:* “Pods may optimize locally.” **Answer:** Evidence Packs include system KPIs (coverage, de-washed GMV) to prevent local maxima.
- *Objection:* “Paperwork overload.” **Answer:** One-pager Evidence Packs; automation for digests; artifacts exist to **accelerate trust**.

Chapter 13 — Annexes (Parameter Registry, SOPs, KPI Specifications)

This chapter aggregates all **machine-actionable** references: on-chain parameters, governance thresholds, operating playbooks (SOPs), data schemas, KPI definitions, threat model, document versions, and glossary. These annexes are designed to be copy-pasted into repos and runbooks without rewording.

Annex A — On-Chain Config Registry (Canonical Keys)

A.1 Registry Contract Surface

- **Program:** `config` (single source of truth; append-only history; timelocked promotions)
- **Events:** `ConfigSet(key, old, new, author, ts)`,
`ConfigPromoted(version, author, proposal_id, eta)`,
`ConfigFrozen(reason)`
- **Access:** updates via `dao.execute` only; no off-ledger toggles

A.2 Parameter Map (keys, types, bounds, rationale)

Key	Type	Bounds / Units	Default	Change Class	Rationale
<code>rewards.burn_bps</code>	u16	0–300 (0–3.00%)	75	Economic	Claim-time discipline; dampens net issuance

<code>rewards.epoch_ceiling_units</code>	u64	≥ 0	1,000,000	Economic	Hard cap per epoch to preserve runway
<code>rewards.max_claim_per_epoch</code>	u64	≥ 0	50,000	Economic	Prevents burst drains
<code>hii.weight.education</code>	u16	0–200	80	Economic	Balance across E/V/A
<code>hii.weight.volunteering</code>	u16	0–200	100	Economic	Default anchor
<code>hii.weight.activity</code>	u16	0–200	60	Economic	Limits low-effort spam
<code>hii.repetition_decay_lambda</code>	u16	0–100	15	Economic	Diminishing returns on repeats
<code>hii.daily_cap_units</code>	u32	≥ 0	100	Economic	Farm-resistant pacing
<code>missions.audit_sampling_bps</code>	u16	0–10000	150	Economic	Random audit baseline (1.5%)

missions.geo_radius_m	u32	10–5000	100	Operational	Proof strength guard
missions.time_window_s	u32	60–86400	3600	Operational	Timestamp tolerance
store.base_fee_bps	u16	0–1000	250	Economic	Marketplace revenue
store.discount_cap_pct	u8	0–50	30	Economic	Subsidy envelope
store.velocity.user.hour	u16	0–100	6	Operational	Anti-wash throttle
store.velocity.merchant.hour	u16	0–1000	60	Operational	Loop prevention
store.settlement_tplus_days	u8	0–14	3 (Std) / 7 (Prob)	Operational	Risk-aligned settlement
buyback.twap_window_s	u32	300–43200	7200	Critical	Anti-manipulation
buyback.oracle_max_age_s	u32	15–300	120	Critical	Price freshness
buyback.coverage_threshold_months	u8	3–24	12	Critical	Runway guard

buyback.max_notional_per_window	u64	≥ 0	100,000	Critical	Slippage control
gov.vote_cap_per_id_bps	u16	1–1000	10	Critical	Anti-capture
gov.quorum.operational_bps	u16	100–5000	1000	Critical	Class quorum (10%)
gov.quorum.economic_bps	u16	1000–10000	3000	Critical	Class quorum (30%)
gov.quorum.critical_bps	u16	3000–10000	6000	Critical	Class quorum (60%)
gov.timelock.operational_s	u32	3600–172800	43200	Critical	$\geq 12h$
gov.timelock.economic_s	u32	43200–604800	172800	Critical	$\geq 48h$
gov.timelock.critical_s	u32	86400–1209600	345600	Critical	$\geq 96h$
sop.emergency_scopes	bitset	Claims/Store/Buyback	0	Critical	Circuit breakers
privacy.retention.mission_days	u16	7–365	90	Operational	“PII-sparse by design”

<code>privacy.retention.dispute_days</code>	u16	7–365	180	Operational	Appeals evidence
<code>compliance.region.flags.*</code>	map	enum	per region	Critical	Jurisdiction toggles

Promotion rule: any change to **Critical** keys requires supermajority + rollback plan + post-change KPI note.

Annex B — Parameter Defaults (v1 Baseline)

- **Burn:** `0.75%` (75 bps)
- **Epoch ceiling:** `1,000,000` reward units
- **Buyback:** **OFF** until (a) realized profit true, (b) coverage ≥ 12 months, (c) oracle fresh
- **HII:** weights $E=0.8$, $V=1.0$, $A=0.6$; repetition decay $\lambda=0.15$; daily cap `100` units
- **Store:** base fee `2.5%`; discount cap `30%`; per-user velocity `6/hour`
- **Governance:** vote cap `0.10%` per ID; timelocks `12h/48h/96h` by class

Annex C — Governance Quorums, Timelocks, Classes

- **Classes:** Operational (low risk), Economic (tokenomics/store/HII levers), Critical (treasury, upgrades, emergency)
- **Quorums:** 10% / 30% / 60% (of eligible power snapshot)
- **Timelocks:** $\geq 12h$ / $\geq 48h$ / $\geq 96h$
- **Single-subject rule:** proposals must map to one intent; bundling is invalid
- **Cooling-off:** parameter flip-flop barred for 7 days unless Emergency SOP invoked

Annex D — SOPs (Runbooks)

Exit: Resume staged; verify dashboard freshness $\leq 60s$; publish a brief post-incident note; store hash on-chain.

User-facing: Banner in MiniApp with status and ETA; disable non-core Store flows; queue claims and display position-in-queue.

Controls: (a) Multi-RPC rotation with health checks; (b) Idempotency keys for submissions/claims; (c) Back-pressure and rate-limiters at the gateway; (d) Delayed governance execution if indexer lag exceeds threshold.

Objectives: Maintain integrity and user communication during L1 congestion or indexer/RPC degradation; prefer queueing over failure.

D.7 Degraded-Mode Operations (Network/Infra Congestion)

6) Rollback: pre-approved rollback path within $\leq 24h$, with breaker conditions documented in the proposal.

5) Reconciliation: compute and publish a Merkle proof of state diffs; publish checksums and receipts for third-party replay.

4) DAO Execute after $\geq 96h$ timelock: freeze old entypoints while keeping state readable; announce ETAs.

3) Canary epoch: route $\leq 1\%$ of production calls to the new program; auto-abort on alarms; archive canary receipts.

2) Read-only parity tests: compare state views, event shapes, and PDA layouts across old vs. new; publish a parity checklist.

1) Shadow deploy the new program with all feature flags off; emit readiness events for observability.

D.6 Upgrade/Migration Runbook (Critical class)

D.1 EmergencyStop (Any Surface)

1. **Trigger** (exploit indicator, stale oracle, regulator order)
2. **Engage** scope flag(s) in `sop.emergency_scopes` via Critical proposal (fast-track path defined)
3. **Notify** status page + MiniApp banner ($\leq 2h$)
4. **Stabilize** (patch, config fix, key rotation)
5. **Resume** staged; monitor KPIs for 72h
6. **Post-mortem** public summary; on-chain hash

D.2 Buyback Execution

- **Pre-checks:** `profit=true`, `coverage≥threshold`, `oracle_age≤max`
- **Queue:** set `twap_window`, `max_notional_per_window`
- **Execute:** DCA fills; immediate **burn** of acquired tokens
- **Report:** emit receipts; include in monthly digest

D.3 Treasury Reconciliation (Monthly)

- Close books (ops costs, fees, partner rev) → compute realized profit → publish digest (hash on-chain) → update coverage months

D.4 Disputes & Appeals

- Intake → evidence request (T+24h) → verdict (≤72h) → log outcome & adjust trust/penalties → anonymized stats in KPI pack

D.5 Security Incident

- Sev triage → contain → forensics → rotate creds/keys → regulator notice if required → 72h post-mortem with actions & owners

Annex E — Data Schemas & KPI Specifications

Feed	Primary	Secondary	Max Age (s)	Fallback Behavior	Abort Rule
\$PEACE/USD	Pyth	Switchboard / Chainlink DON	≤120	Freeze buybacks; retry on fresh tick	Abort if age>120 or dev > X bps
SOL/USD	Pyth	Switchboard / Chainlink DON	≤120	Halt buyback calculations	Abort if age>120 or dev > X bps
Store FX (optional)	Trusted FX (Pyth)	Switchboard	≤300	Fallback to last-good; annotate digest	Abort category if deviation beyond policy

E.0 Oracle Providers & Max-Age (Source of Truth)

Oracle Providers & Max-Age” table (Chainlink/Pyth/Switchboard; $\leq 120s$)

E.1 Canonical Tables (minimal columns)

- `missions_verified(id, user, type, host, ts, verdict, proof_hash)`
- `rewards_claimed(tx, user, epoch, gross, burn_bps, net, ts, config_version)`
- `store_orders(id, user, merchant, amount, discount_pct, flagged, settled, ts)`
- `buybacks_executed(id, twap_window_s, notional, oracle_age_s, ts)`
- `gov_votes(proposal_id, user, class, weight, ts)`
- `circulating_supply(ts, amount)`

E.2 KPI Definitions (auditable math)

- **MVR (Mission Verification Rate)** = `count(verified)/count(submitted)`
- **TVDR (Token Velocity Decay Rate)** = $\Delta(1/velocity)$ over rolling window (30/90d)
- **Coverage Months** = `(treasury_free_cash / rewards_run_rate_monthly)`
- **Clean GMV** = `GMV - sum(flagged_value)` (after de-wash outcomes)
- **De-wash Rate** = `sum(flagged_value)/GMV`
- **Repeat-Merchant Rate** = `merchants(active 3 consecutive months)/merchants(active)`
- **Governance Herfindahl** on effective voting power post-caps/HII
- **ERS (Ecosystem Resilience Score)** = normalized composite of Coverage, GMV stability (post de-wash), stake/lock ratios

E.3 Publishing Cadence

- **Near-real-time tiles** ($\leq 60s$ lag)
- **Monthly Signed Digest:** CSV/Parquet + query fingerprints + on-chain hash

Annex F — Threat Model (Summary)

Class	Attack	Primary Controls	Secondary Controls
Sybil	Multi-accounting	Soulbound Passport; device KYC tiers; duplicate detection heuristics	
Proof forgery	GPS/media spoof	Geofence & dwell; watermarking	Random audits; host reputation
Store wash	Loop GMV	Velocity caps; loop detection	Tiered settlement; dispute reserves
Governance capture	Vote concentration	Per-ID cap; HII weight	Quadratic dampening; class quorums
Oracle manipulation	Stale/thin prices	Oracle freshness; TWAP	Per-window notional cap; circuit breaker
Treasury misuse	Optics buybacks	Profit-only; coverage threshold	Public receipts; digest reconciliation
Data exposure	PII breach	PII off-chain; encryption; RBAC	DPIA/ROPA; deletion paths

Annex G — Document Versions (External-Facing)

- **Terms of Use v1.0** — jurisdiction/venue, acceptable use, disclaimers
- **Privacy Notice v1.0** — purposes, retention, rights, contacts
- **Merchant Agreement v1.0** — fees, SLA, reserves, category rules
- **Open Methodology v1.0** — KPI math, Clean GMV, buyback SOP excerpt
- **Risk Factors v1.0** — operational, market, legal, technology, governance

All documents semantically versioned; release notes and hashes recorded publicly.

Annex H — Glossary (Selected)

- **HII** — Human Impact Index (score weighting E/V/A missions with decay, caps, and verification status)
- **Clean GMV** — GMV net of flagged/invalidated orders post de-wash pipeline
- **Coverage Months** — reward runway at current accrual rate and parameters
- **TWAP** — Time-Weighted Average Price; anti-manipulation execution window
- **Config Registry** — on-chain parameter store with versioning and timelocks
- **Evidence Pack / Signed Digest** — public KPI bundle with on-chain content hash

End Matter — Consolidated Actionable Recommendations & Appendices

EM-1 — Consolidated Actionable Recommendations

EM-1.1 Governance & Registry

- Freeze **Critical** keys until post-beta: upgrades, buyback toggles, oracle age, coverage threshold.
- Promote **Config v1** with defaults (Annex B) and publish diff + ETA.
- Stand up **proposal hygiene**: single-subject checklists, disclosure templates, rollback boilerplates.

EM-1.2 Tokenomics & Treasury

- Set **burn_bps=75** and **epoch_ceiling_units=1,000,000**; instrument **Coverage Months** tile.
- Keep buybacks **OFF**; run in **monitor** mode for 60 days to baseline profits and liquidity; draft venues list.
- Publish circulating supply & reward coverage tiles; link fingerprints in the first digest.

EM-1.3 HII & Missions

- Launch **3 baseline templates** (E/V/A) with conservative points; enable **audit_sampling_bps=150**.
- Configure repetition decay $\lambda=0.15$; daily caps at **100** units; appeals SLA $\leq 72h$.
- Start host reputation scoring; schedule first **random site audits** within 30 days.

EM-1.4 Store & Commerce

- Onboard **20 probation merchants** across low-risk categories; set per-user velocity **6/hour**, discount cap **30%**.
- Enable de-wash pipeline (loop detection + velocity anomalies) **before** first promo.
- Define **Net Contribution floor** per category; auto-downshift discount elasticity if breached.

EM-1.5 Data & Observability

- Ship indexer **MVP** with canonical tables (Annex E) and **query fingerprints**.
- Publish public dashboard: MVR, Claims, Clean GMV, Coverage, TVDR, Governance Herfindahl.
- Generate **Monthly Signed Digest** (CSV/Parquet + on-chain hash) starting Month 1.

EM-1.6 Security, Privacy, Compliance

- Complete **DPIA v1**; implement deletion/erasure workflow; verify no PII in logs.
- Run **EmergencyStop** tabletop; document outcomes; store hash on-chain.
- Stand up sanctions/KYC regional toggles but leave stricter tiers **off** until jurisdiction mapping finalizes.

EM-1.7 Organization & Delivery

- Staff minimum **pods & councils** (Chapter 12); assign DRIs; publish on-call rotations.
- Adopt **dual-lane** release train (Value + Proof); block launches without Evidence Packs.
- Schedule external **security audit** slot for the end of Open Beta.

EM-2 — Appendices (Chapter-Scoped Reference Packs)

EM-2.1 Chapter 1 (Executive Summary & Introduction) — Appendix

- **Flywheel diagram spec** (Impact → Value → Voice → Stability)
- **Module–Control Surface map** (Access, Identity, Proof, Incentives, Commerce, Governance → Registry keys & SOP links)
- **Principles memo** (utility-first token; profit-conditioned buybacks; privacy by design)

EM-2.2 Chapter 2 (Vision) — Appendix

- **Constellation framework** one-pager (A–F “stars”)
- **North-Star KPIs** (adoption, proof quality, solvency, merchant gravity, governance health)
- **12-month KPI targets** and visualization mocks

EM-2.3 Chapter 3 (Market Need) — Appendix

- **Trust Supply Chain Index (TSCI)** methodology (weights, sampling)
- **Regional readiness checklist** (rails, regulation, NGO density, merchant mix)
- **Benchmark sources** list skeleton for ongoing updates

EM-2.4 Chapter 4 (Architecture) — Appendix

- **Systems map** (client, gateway, programs, indexer)
- **Config Registry key taxonomy** (pointer to Annex A)
- **Interface stubs** for `passport`, `missions`, `rewards`, `treasury`, `store`, `dao`, `config`

EM-2.5 Chapter 5 (Technical Specs) — Appendix

- **Event payload contracts** (selected JSON examples)
- **Compute budget targets** per hot path

- **Test matrix** (unit/prop/fuzz; invariants; soak)

EM-2.6 Chapter 6 (Tokenomics) — Appendix

- **Emission calculator** worksheet spec (inputs: burn bps, epoch ceiling, adoption scenarios)
- **TVDR methodology** and interpretation guide
- **Buyback SOP excerpt** (eligibility gates & receipts fields)

EM-2.7 Chapter 7 (Governance) — Appendix

- **RAPID/RACI** sheet templates
- **Disclosure pack** template for proposals (summary, risk, diffs, rollback)
- **Delegate dashboard** metrics (turnout, concentration, performance)

EM-2.8 Chapter 8 (HII & Missions) — Appendix

- **Template library** (E/V/A v1 specs)
- **Scoring function** parameters cheat-sheet (α , λ , caps)
- **Appeals playbook** (roles, SLA, evidence checklist)

EM-2.9 Chapter 9 (Peace Store) — Appendix

- **Category policy** table (risk tiers, ceilings, settlement schedules)
- **De-wash pipeline** decision tree (flags → outcomes)
- **Net Contribution** formula examples by category

EM-2.10 Chapter 10 (Risk/Legal/Privacy) — Appendix

- **Compliance matrix** (controls → evidence) snapshot
- **Incident comms** templates (user, partner, regulator)

- **Data retention** schedule tables

EM-2.11 Chapter 11 (Roadmap & Delivery) — Appendix

- **Phase exit criteria** one-pager per phase
- **Evidence Pack** template (OKRs, tiles, fingerprints, summary)
- **Roll-back criteria** decision tree

EM-2.12 Chapter 12 (Team & Org) — Appendix

- **Org chart** v1
- **Hiring plan** Gantt blocks and leveling guide
- **Performance rubric** (promotion signals & anti-patterns)

Store anti-wash velocity caps: user 6/hr, merchant 60/hr; discount cap 30%; base fee 2.5%.

Peace Constellation (Positioning)

A concise metaphor aligning Passport (North Star), Missions (planets), HII (constellation lines), \$PEACE (fuel), the Peace Store (habitable zones), and DAO governance (gravitational laws). The flywheel operates as: Impact → Verification → Rewards → Commerce → Governance → Parameter updates—keeping orbit via transparent, code-enforced guardrails.

Impact Gravity Index (IGI)

IGI measures ecosystem “pull” by tying verified impact to incentive load.

Formula: $IGI = (\text{Verified Missions} \times \text{HII Quality}) / (\text{Reward Claims} + \text{Store Subsidy})$.

Targets: IGI > 0.8 healthy; IGI < 0.6 indicates farming pressure; IGI > 1.2 suggests over-verification friction.

Governance Health Score (GHS)

$GHS = \text{turnout_rate} \times (1 - \text{Herfindahl_index})$ on effective voting power (after caps and optional HII-weight).

Target > 0.7. If GHS < 0.5, auto-pause treasury proposals until recovery is confirmed by two consecutive voting cycles.

Risk Absorption Ratio (RAR)

$RAR = \text{mitigations} / \text{risk_vectors with target} > 0.8.$

Governance must maintain ≥ 2 orthogonal controls per high-severity vector; publish RAR and changes in monthly digests.

Verification Pipeline — Signals & Weights

Default signal weights: GPS 0.3, Media 0.4, Attestation 0.2, Behavior 0.1. Auto-verify target $\geq 95\%$ after calibration (initial beta target 85–90%) via ZK (Groth16) + oracle corroboration; 5% risk-based audits.

Figures (All figures will be injected into the final version)

(Figures will be inserted into the final Doc after PS brand Identity created and approved)
(The Sample Figures are just illustration and-not final)

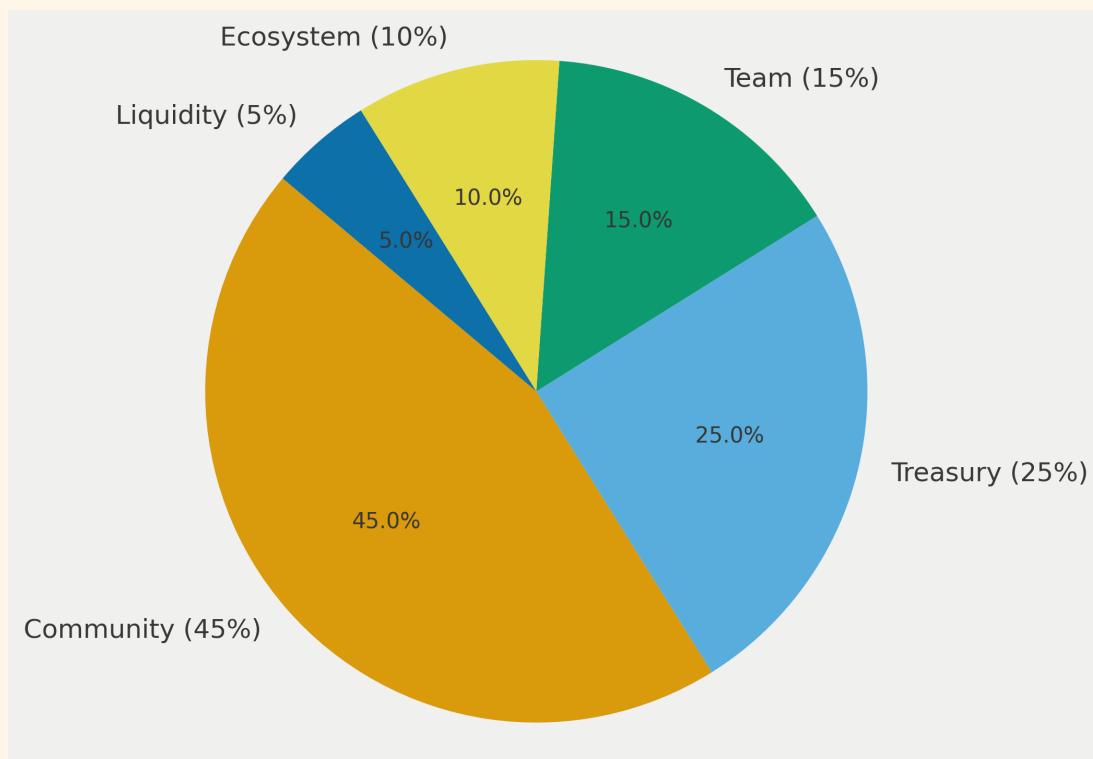
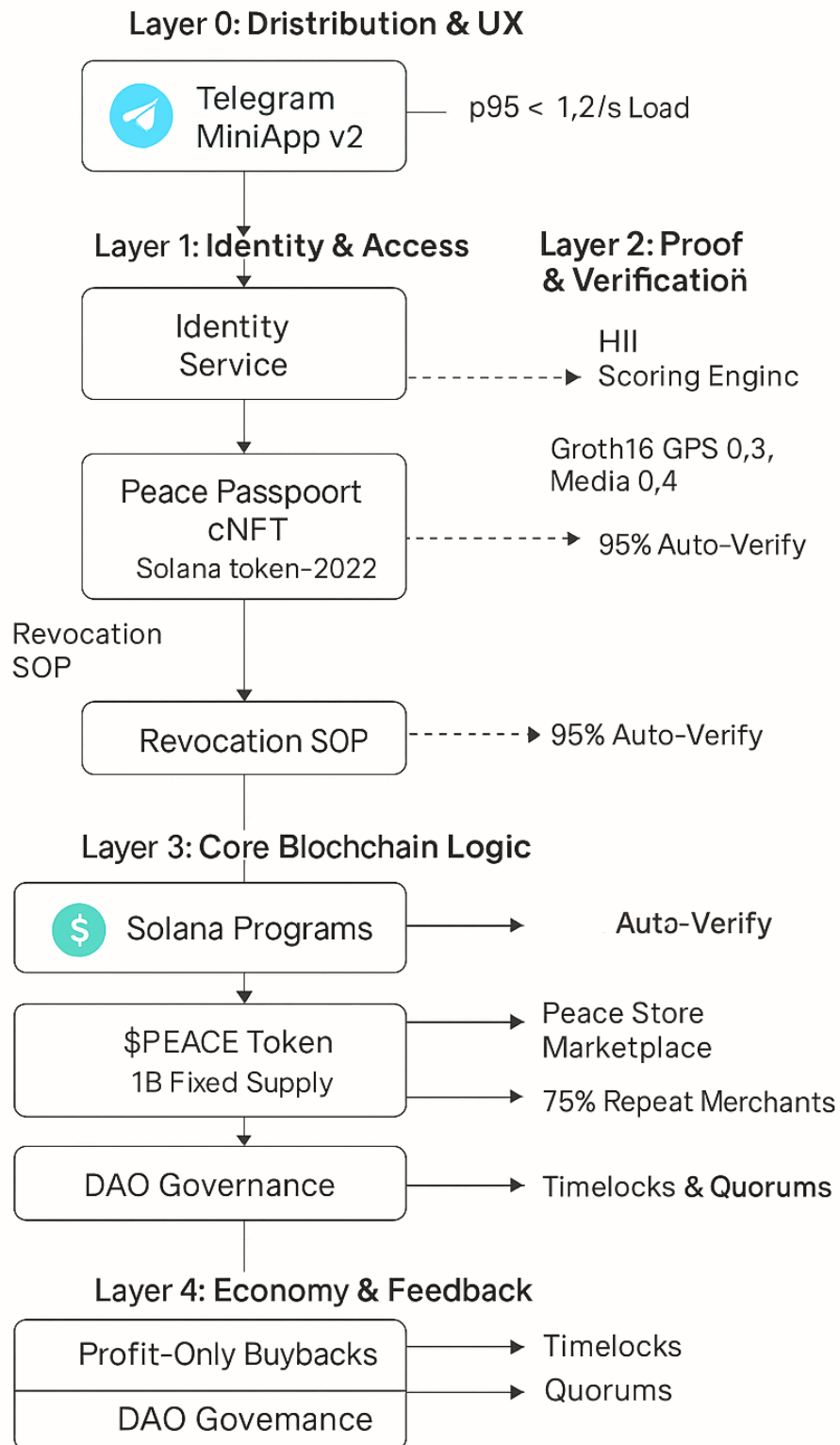


Figure 1. Token Allocation (illustrative): Community 45%, Treasury 25%, Team 15%, Ecosystem 10%, Liquidity 5%

Peace Soldiers Layered Architecture



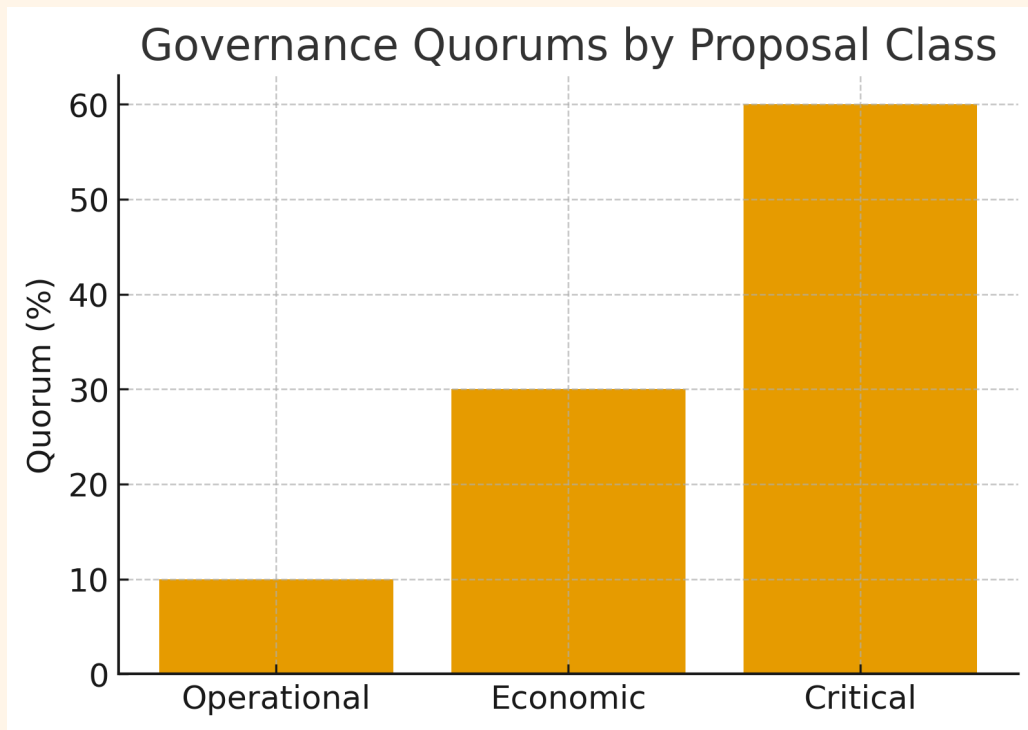


Figure 2A. Governance Quorums by Proposal Class: Operational 10%, Economic 30%, Critical 60%.

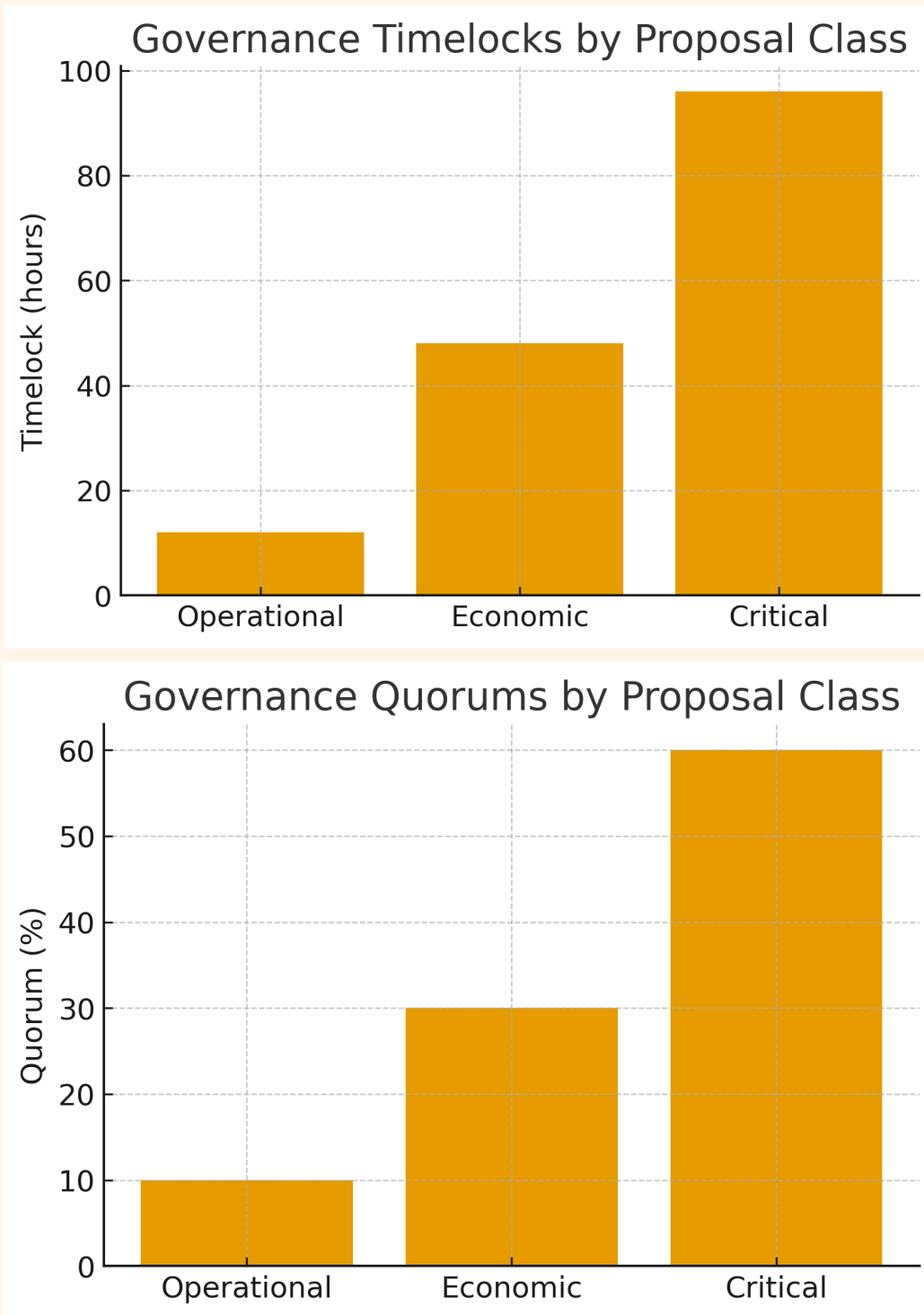
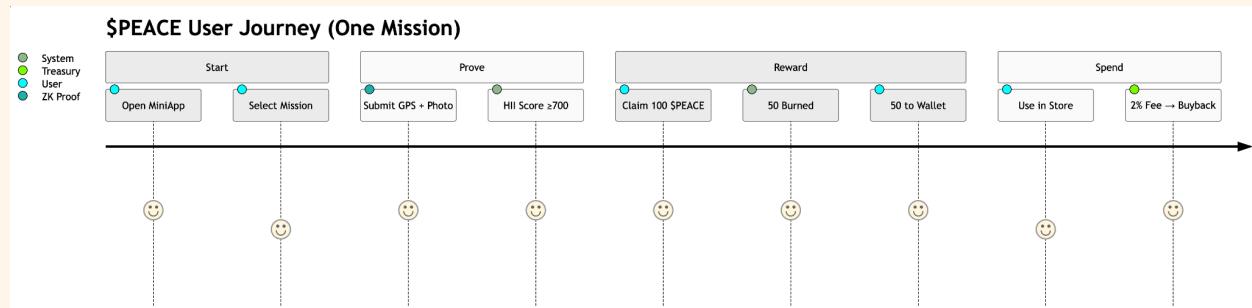


Figure 2B. Governance Timelocks by Proposal Class: Operational 12h, Economic 48h, Critical 96h.



References

- [R1] Solana Docs — Transaction Fees. <https://solana.com/docs/core/fees>
- [R1a] Solana Learn — Understanding Transaction Fees.
<https://solana.com/learn/understanding-solana-transaction-fees>
- [R2] TechCrunch — Telegram hits 1B MAUs (Mar 19, 2025).
<https://techcrunch.com/2025/03/19/telegram-founder-pavel-durov-says-app-now-has-1b-users-calls-whatsapp-a-cheap-watered-down-imitation/>
- [R3] Business Insider — Telegram 1B MAUs (Mar 2025).
<https://www.businessinsider.com/telegram-celebrates-1-billion-active-users-pavel-durov-swipe-whatsapp-2025-3>
- [R4] Telegram Dev Docs — Mini Apps. <https://core.telegram.org/bots/webapps>
- [R5] Chainalysis — 2025 Crypto Crime Report (Intro).
<https://www.chainalysis.com/blog/2025-crypto-crime-report-introduction/>
- [R6] Chainalysis — 2025 Crypto Crime Report (PDF).
<https://go.chainalysis.com/2025-Crypto-Crime-Report.html>
- [R7] Chainlink 2.0 Whitepaper — Decentralized Oracle Networks. <https://chain.link/whitepaper>
- [R8] Chainlink Docs — Overview. <https://docs.chain.link/>